SEDGM: A Structure-Enhanced Spatial–Spectral Dynamic Gating Mamba for Hyperspectral Image Classification

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***Abstract*—With the rapid development of hyperspectral image classification (HSIC) technology, its applications in geological exploration and environmental monitoring have become increasingly prominent. Recently, Mamba has garnered significant attention owing to its outstanding performance in long-range sequence modeling and linear computational complexity. However, Mamba still exhibits significant limitations in HSIC: first, it does not fully consider the hierarchical spatial–contextual representation and nonlinear spectral interactions in hyperspectral images; second, its sequential processing approach leads to the loss of spatial structural information and feature redundancy. In response, this study proposes a structure-enhanced spatial–spectral dynamic gating Mamba (SEDGM) that leverages the collaborative design of spatial and spectral gating Mamba mechanisms to extract and exploit key regional features of hyperspectral data. The spatial branch employs hierarchical gating Mamba (HGM) to capture multidirectional pixel sequences and extract the hierarchical spatial features and their intrinsic relationships. In contrast, the spectral branch utilizes a random shuffled gating Mamba to disrupt the fixed order of traditional spectral sequences and capture higher order spectral couplings, effectively characterizing the cooperative variation patterns of spectral features. Both branches employ a dynamic gating mechanism that weights features based on sequence centrality, dynamically activating feature sequences. Additionally, shape-specific offset-aware attention (OAA) is incorporated into each branch to enhance the structured features that were lacking in the Mamba sequences. Finally, a spectral-oriented feature review module (SOFRM) is incorporated to achieve dynamic feature fusion and optimized refinement. Experiments were conducted on four large-scale benchmark hyperspectral imaging (HSI) datasets, with SEDGM achieving significant improvements in classification performance, validating the effectiveness of this approach in HSIC tasks. The code is available at https://github.com/shuai2023-hash/SEDGM**

Received 24 June 2025; revised 9 August 2025 and 22 September 2025; accepted 22 October 2025. Date of publication 30 October 2025; date of current version 17 November 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42301424 and Grant 42171341. *(Corresponding author: Shuai Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3626930

***Index Terms*—Feature review, gating Mamba, hyperspectral imaging (HSI) classification, offset-aware, structure-enhanced.**

I. Introduction

HYPERSPECTRAL imaging (HSI) has become a crucial technical means for Earth observation and analysis. Compared with conventional optical red, green, blue (RGB) images, HSI possesses broader spectral coverage and narrower spectral channels, with imaging bands ranging from dozens to hundreds, spanning visible to far-infrared wavelengths. This high spectral resolution ensures that each pixel contains rich spectral information, significantly enhancing the discriminative capability for similar ground objects [1], [2]. Consequently, HSI has been widely applied in geological exploration, environmental monitoring, urban mapping, precision agriculture, and other fields [3], [4]. In these applications, HSI classification serves as a fundamental step, with its core task being the accurate assignment of category labels to each pixel [5].

With the rapid development of deep learning techniques, significant progress has been made in the field of HSI classification. Current mainstream approaches primarily include frameworks based on convolutional neural networks (CNN) [6], graph convolutional networks (GCNs) [7], and transformers [8]. CNN-based methods, leveraging their advantages of local connectivity and weight sharing, have demonstrated powerful local feature extraction capabilities and computational efficiency in hyperspectral image processing, as exemplified by 3-D CNN [9], LWCNN [10], HybridSN [11], and SSRN [12]. The introduction of attention mechanisms has further enhanced CNN performance, such as Zhu et al. [13] proposed RSSAN, which employs spectral and spatial attention modules to achieve dynamic feature weighting. However, the fixed receptive field of CNNs limits their global context modeling capability. Although increasing network depth can expand the receptive field, it leads to substantially higher computational costs and struggles to adapt to geometric deformations of complex ground objects [14]. GCN attempts to overcome local perspective limitations by using graph structures as input, processing graph-structured data through constructed node relationships to simulate long-range dependencies in images. Examples include Shahraki and Prasad [15] constructing semisupervised adjacency matrices that fuse labeled and unlabeled data, Qin et al. [16] developing second-order

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GCNs to simultaneously capture spatial–spectral neighborhood relationships, and Wan et al. [17] optimizing graph structure inputs using superpixel segmentation. Although such methods can simulate long-range dependencies, the construction of full-graph adjacency matrices results in high computational complexity, severely restricting their application in large-scale scenarios [18]. Transformer-based methods achieve global context modeling through self-attention mechanisms. For instance, Yang et al. [19] proposed HiT, which enhances feature extraction capability via spectral-adaptive 3-D convolution and convolutional transformer modules, while Roy et al. [20] integrate morphological convolution with self-attention mechanisms to optimize feature representation through spectral–spatial morphological operations. Although such approaches can establish global relationships between pixels, the self-attention mechanism exhibits extremely high computational complexity when processing high-dimensional spectral data. Moreover, the linear modeling approach struggles to capture nonlinear relationships between spectral bands, and the massive model parameters increase the risk of overfitting [21].

State space models (SSMs) have attracted attention due to their long-range modeling capability with linear complexity. Among them, Mamba achieves input adaptation through dynamic selection mechanisms [22]. When processing diverse input data types and complexities, Mamba can dynamically adjust its internal parameters and structure to maintain stable performance across different scenarios. This adaptive characteristic gives Mamba significant advantages in complex tasks like HSI classification, enabling more accurate identification of different class features while effectively improving classification accuracy and efficiency [23]. Benefiting from these advantages, methods such as SpectralMamba [24] and MambaHSI [25] were the first to introduce Mamba into the hyperspectral domain. By performing selective scanning of 3-D hyperspectral tokens in both spatial-spectral feature space and hidden state space, they achieve simplified yet sufficient dynamic modeling of HSI data, thereby extracting global spectral–spatial semantic representations. And S2Mamba further incorporates patch interaction and bidirectional spectral scanning to enhance contextual dependencies [26].

Despite these advancements, they still primarily focus on improving scanning strategies, without fully addressing Mamba's fundamental challenges in HSI classification. Current Mamba-based hyperspectral classification methods still exhibit the following three limitations.

1) *Spatial Feature Limitation:* Ground objects in HSI vary in size, exhibit diverse morphologies, and have complex boundaries. Single-scale sequential scanning struggles to construct effective hierarchical spatial features, leading to a trade-off in these multiscale scenarios: small receptive fields cannot grasp the overall semantics of large ground objects, while large receptive fields blur the fine boundaries of small objects, as shown in Fig. 1(a).
2) *Spectral Modeling Limitation:* Current spectral scanning methods rely on linear sequential scanning, failing to model nonlinear relationships between bands, which limits the model's spectral discriminative power. Furthermore, adjacent hyperspectral bands often exhibit high similarity; directly utilizing these redundant features reduces interclass separability, as shown in Fig. 1(b).
3) *Structural Information Limitation:* In hyperspectral classification, the class of a central pixel strongly depends on its neighborhood information. However, Mamba's sequential processing breaks spatial topology, separating adjacent pixels in the sequence. This disrupts texture continuity and shape constraints, making structural information difficult to utilize effectively, as shown in Fig. 1(c).

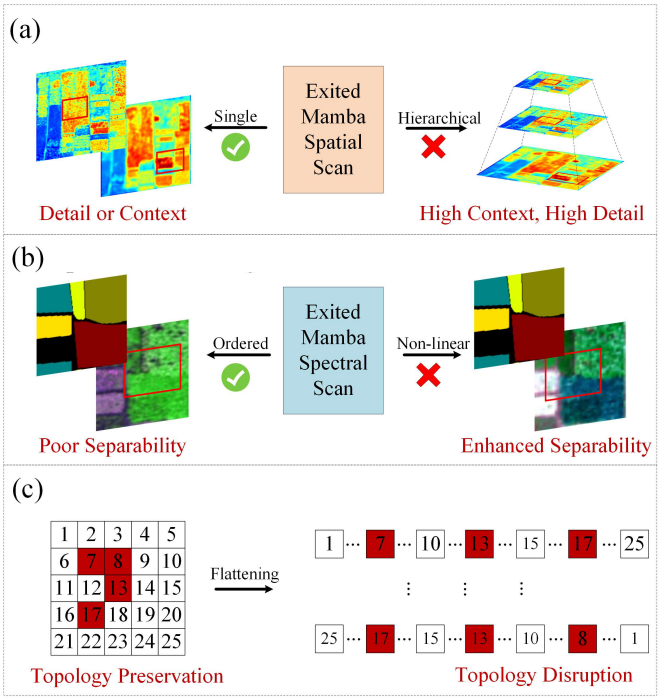


Fig. 1. Limitations of current Mamba-based hyperspectral classification approaches. (a) Spatial Feature Limitation: Single-scale receptive fields struggle to simultaneously capture fine details and global context, leading to a lack of hierarchical local features. (b) Spectral Modeling Limitation: Solely relies on sequential scanning, neglecting nonlinear combinations between spectral bands. (c) Structural Information Limitation: Sequential processing disrupts spatial topological relationships, leading to the loss of texture continuity and shape constraints.

To address these issues, we proposes the SEDGM, which flexibly adapts to diverse data patterns while effectively capturing long-range spatial correlations and spectral dependencies. The model employs a hierarchical gating Mamba (HGM) and a random shuffle gating Mamba (RSGM) to extract hierarchical spatial features and nonlinear spectral features, respectively. Additionally, we introduce two variants of offset-aware attention (OAA) mechanisms within both scanning blocks to enhance the structural features typically lacking in Mamba's sequential data processing. Finally, a specially designed spectral-oriented feature review module (SOFRM) is introduced to dynamically fuse and review spectral–spatial features by introducing spectral characteristics and gating-KAN.

The contributions of this study are as follows.

1) We propose the HGM and RSGM. HGM performs multidirectional scanning to extract hierarchical spatial

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features and their relationships. RSGM disrupts traditional spectral sequences through random shuffling to capture higher order spectral couplings, enhancing feature discriminability. Both modules employ a dynamic gating mechanism that weights features based on sequence centrality, dynamically activating feature sequences.

2) We present the OAA by integrating multihead attention with spatial sampling techniques. By introducing sampling offsets, OAA enables flexible attention allocation across spatial locations, significantly enhancing the model's spatial perception capability. This approach allows the model to dynamically adjust the focus area over a broader range, concentrating on relevant regions and capturing key features.
3) We propose SOFRM that introduces spectral information as dynamic conditioning factors. This module establishes adaptive mapping relationships between spectral and spatial features, enabling the multiscale fusion of nonlinear spectral characteristics with spatial features, and incorporates Gate-KAN to selectively reinforce learned features while acquiring novel high-order representations.
4) This study conducts extensive experiments on four datasets, exhibiting the superior classification performance of the proposed method.

The remainder of this article is organized as follows. Section II reviews studies related to HIS classification. Section III provides a detailed description of the proposed SEDGM. Section IV describes the datasets and analysis of the experimental results. Section V discusses the ablation studies. Finally, Section VI presents the conclusions.

II. Related Work

This section provides a comprehensive review of deep learning-based HSIC methods.

A. HSIC Methods

CNNs have garnered significant attention in the field of HSIC because of their exceptional feature extraction capabilities. Wang et al. [27] enhanced information flow using dense connections and convolutional dimensionality reduction, boosting classification efficiency. Zhong et al. [12] developed consecutive 3-D CNN modules to extract discriminative spectral–spatial features from HSI data cubes. Yu et al. [28] introduced an attention CNN with feedback mechanisms to optimize feature representation. To address computational complexity, Li et al. [29] proposed a full-group CNN employing group convolution and shuffle operations. Zhang et al. [30] designed an evolutionary CNN that adaptively optimizes network parameters through multiobjective search. Afjal et al. [31] combined 3-D–2-D CNNs with multibranch fusion and Seg-PCA for improved classification. However, the limitation of CNNs lies in their limited receptive fields, which make it difficult to capture global contextual information effectively. Although increasing the depth of the network can expand the receptive field, it significantly increases the computational burden, leading to a trade-off between efficiency and performance. Additionally, when faced with complex and varied land-cover types and geometric shape differences, CNNs exhibit shortcomings in capturing geometric variations between regions, particularly in class-boundary areas where critical information is easily lost, thereby limiting their classification accuracy.

GCNs have provided a new research direction for HSIC owing to their advantages in processing unstructured data. Sha et al. [32] proposed a semisupervised framework using graph attention networks (GATs) that dynamically weights neighboring nodes to generate enhanced embeddings. Ding et al. [33] developed a multiscale graph network that learns hierarchical features while emphasizing key spatial objects through GAT. Xu et al. [34] introduced a hypergraph convolutional network (HGCN) that captures global and local features via hyperedge construction. Duan et al. [35] extended this work by combining standard convolution with hypergraph convolution in their SPHGCN model to better model spatial–spectral relationships. Ding et al. [36] applied an adaptive smoothing technique based on Kullback–Leibler divergence to reduce spectral variability while preserving node diversity. Chen et al. [37] designed a spectral–spatial dynamic GCN (SSD-GCN) that adjusts graph structures based on node similarity and spectral–spatial features. Although GCN methods attempt to overcome the limitations of local perspectives through global graph structure inputs, their demand for computational resources remains substantial, which restricts their application in large-scale datasets and real-time scenarios.

Transformer-based approaches have demonstrated strong potential in HSIC by effectively modeling long-range dependencies. Gao et al. [38] developed a spatial–spectral vision transformer (SSViT) that processes HSI data through linear mapping of spatial and spectral sequences. Wu et al. [39] improved vision transformers by incorporating convolutional operations, enhancing both performance and efficiency. SpectralFormer [40] rethought HSIC from a temporal perspective, utilizing adjacent bands to learn local spectral sequence information; however, this approach overlooked spatial location information. To address this, SSFTT and CASST adopt a spatial–spectral dual-branch transformer, capturing both spectral and spatial information through a Gaussian-weighted feature tokenizer and a weighted shared spatial–spectral cross-attention module, respectively [41], [42]. Xu et al. [43] proposed the CS2DT, which combines an adaptive dense encoder and a cross-spatial–spectral attention module to achieve efficient extraction and fusion of spatial–spectral features. Yang et al. [44] designed a GTFN that enhanced classification performance by combining contextual information and long-range dependencies in the spectral domain. However, transformers tend to overlook the local features when processing global dependencies. To address this issue, Qi et al. [45] embedded 3-D convolutions within transformers and Song et al. [46] proposed a BS2T and constructed a dual-branch framework combined with a 3-D CNN to jointly extract the local–global features of HSI. Zhao et al. [47] designed a GSC module that significantly reduced the number of parameters and effectively captured local spectral–spatial

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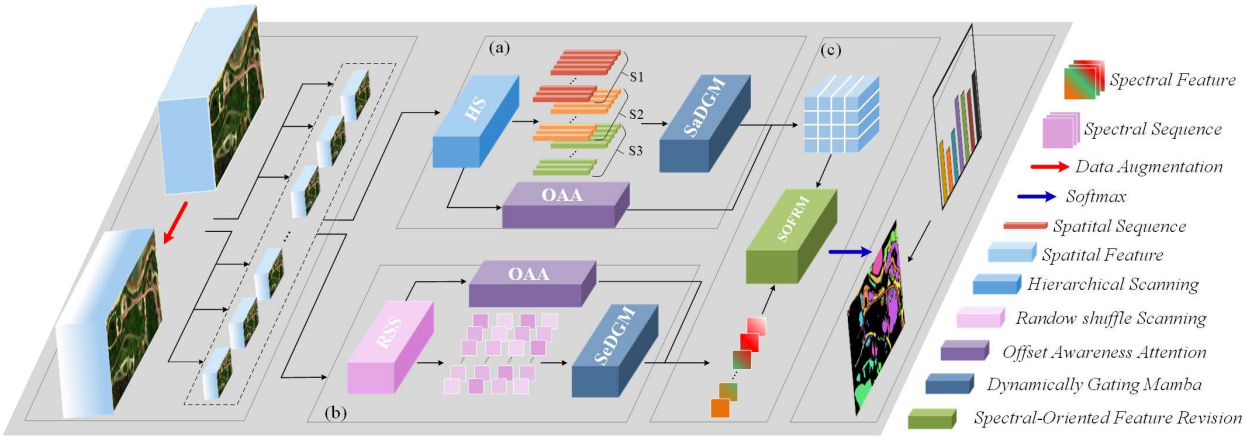


Fig. 2. Overall structure of the proposed SEDGM, which consists of four parts: HGM, RSGM, OAA, and SOFRM, where gray boxes delineate the model's distinct functional components: the leftmost box represents the preprocessing stage, the central section sequentially contains core modules including HGM, RSGM, OAA, and SOFRM, while the rightmost box corresponds to the classification output stage. Furthermore, the OAA is extended to operate in both spatial and spectral dimensions, with its implementations strategically integrated into the HGM and RSGM, respectively. (a) HGM. (b) RSGM. (c) SOFRM.

information in an HSI using GSC-ViT. However, these methods do not fully consider the multiscale differences in ground objects in remote sensing images, which limits their ability to recognize multiscale ground objects. Yang et al. [48] proposed interactive learning frameworks, and Xue et al. [49] implemented spatial partition recovery modules. Li et al. [50] proposed a lightweight transformer model called SPFormer, which reduces the number of parameters and is suitable for few-shot HSIC. However, owing to its quadratic complexity, the self-attention mechanism in transformers faces challenges in terms of computational speed and memory usage when processing high-dimensional hyperspectral data. Additionally, modeling the long-range dependencies of spectral–spatial features requires a large amount of labeled data to learn spectral–spatial features, leading to a higher risk of overfitting.

B. Mamba-Based Methods

Recently, SSMs have garnered significant attention owing to their ability to effectively model long-range dependencies while maintaining linear complexity and have gradually emerged as a novel alternative to CNNs and transformers. Building on SSMs, Mamba incorporates a selection mechanism that allows it to focus adaptively on input components, demonstrating promising results in visual tasks. Ma et al. [51] designed a hybrid CNN-SSM block by combining the local feature extraction capability of convolutional layers with the ability of SSMs to capture long-range dependencies. Wang et al. [52] proposed a novel architecture, Mamba-UNet, that synergizes the capabilities of U-Net and Mamba for medical image segmentation, capturing intricate details and broader semantic contextual information in medical images. Zhu et al. [53] proposed a novel vision backbone network based on a bidirectional Mamba block (ViM), which compresses visual representations through position embedding and bidirectional state space models, demonstrating that visual representation learning does not rely on self-attention mechanisms. In the hyperspectral domain, SpectralMamba proposes an efficient and effective deep learning framework integrating SSMs for HSI classification based on both pixelwise and patchwise inputs. By leveraging simplified yet sufficient dynamic modeling of HSI data in both spatial–spectral feature space and hidden state space, the framework effectively mitigates the impacts of spectral variability and spectral confusion [24]. However, these methods focus on spectral information, not interacting with spatial information. MambaHSI presents an end-to-end framework featuring a spatial–spectral mamba block (SpeMB) that separately processes spatial and spectral features. With residual connections and SSM's global modeling capacity at linear complexity, it outperforms CNNs and transformers in long-range dependency modeling while remaining computationally efficient [25]. Wang et al. [26] designed two selective structured state space models of different dimensions in s^2mamba for spatial and spectral feature extraction and developed a spatial–spectral hybrid gate for optimal fusion. Huang et al. [54] designed SS-Mamba with cooperative spectral–spatial tokens. Sheng et al. [55] developed a parallel lightweight Mamba and CNN block to extract global and local spectral–spatial features, incorporating dynamic position embedding in the network to enhance the spatial location information of visual sequences and further improve the computational efficiency of the Mamba structure. He et al. [56] designed a 3DSSMamba framework tailored to the characteristics of hyperspectral data by introducing a 3DSS mechanism. This mechanism enables pixel-by-pixel selective scanning of 3-D hyperspectral tokens along the spectral and spatial dimensions, thereby achieving the extraction of global spectral–spatial semantic representations. Zhou et al. [57] proposed an innovative mamba-in-mamba (MiM) architecture that includes an MCS mechanism, a T-Mamba encoder with an STL and STF, and a WMF module, significantly enhancing feature generation, centralization, and decoding efficiency. However, there are some limitations to Mamba-based models for HSI classification. First, exited mamba-based models tend to adopt a unidirectional scanning strategy, which fails to fully consider the complex nonlinear relationships between spectral dimensions. Second, Mamba employs a single-

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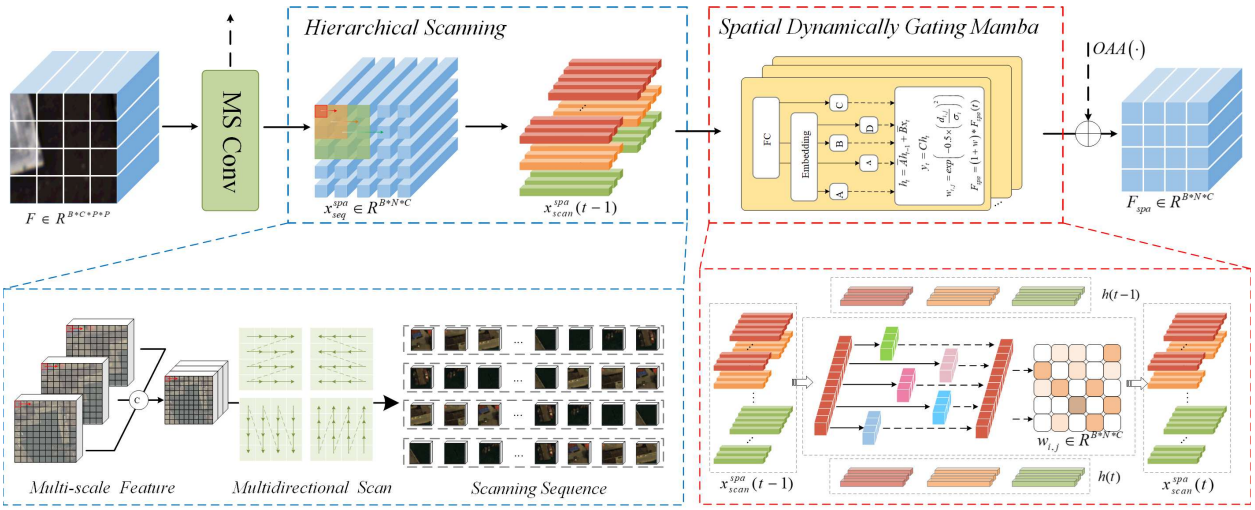


Fig. 3. Overall structure of HGM, which includes hierarchical scanning and spatial dynamic gating Mamba. Here, $F \in R^{B*C*P*P}$ and $F_{\text{spa}} \in R^{B*N*C}$ represent the input and output features of HGM, respectively, $x^{\text{spa}}_{\text{scan}}(t-1)$ and $h(t-1)$ denote the input sequence and state variables of the Gate Mamba, and $w_{i,j} \in R^{B*N*C}$ represents the dynamic gating weights.

scale scanning approach, which limits its ability to capture information regarding ground objects of different sizes. Finally, Mamba treats data as sequences for processing. Although this mode has advantages in handling sequential data, it overlooks the structural information that the data may contain. For hyperspectral imagery, the spatial structures and relationships between pixels and regions are crucial for accurate understanding and classification.

III. Proposed Method

This section outlines the proposed SEDGM, which captures spatial and spectral features through HGM and RSGM, respectively, and utilizes the SOFRM for fusion, as illustrated in Fig. 2. HGM generates pixel sequences at different viewing distances and performs multidirectional scanning to thoroughly mine the hierarchical spatial features. The RSGM captures more discriminative spectral features by introducing shuffled paths. Additionally, we incorporate OAA into the scanning blocks to enhance the structured features of the sequence data. Finally, the SOFRM dynamically enhances and reviews the fused features.

A. Hierarchical Gating Mamba

Traditional Mamba processes hyperspectral data as a 1-D sequence in a unidirectional manner. Although this paradigm performs well in temporal data modeling, it struggles to effectively capture the inherent hierarchical spatial structure characteristics of hyperspectral images and preserve the spatial topological relationships of the image data. However, in hyperspectral remote sensing image classification tasks, the full utilization of multiscale features is crucial for accurately extracting and interpreting complex spatial information. Inspired by this, we designed an HGM as shown in Fig. 3. First, HGM employs a multiscale, multidirectional scanning strategy to effectively retain multilevel structural information ranging from local textures to global spatial features.

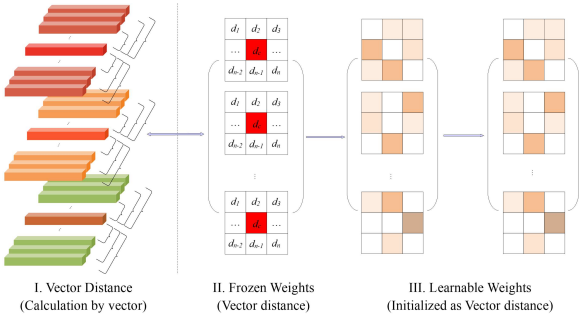


Fig. 4. Learning process of dynamic gating weights in HGM. (a) Stage I: vector-distance computation. (b) Stage II: frozen gate weights (from vector distance). (c) Stage III: learnable gate weights (initialized by vector distance).

Specifically, we implement an equivalent scale transformation in the receptive field domain using multiscale convolution operations, which addresses two critical limitations of physical transformation methods: 1) spatial information loss caused by downsampling operations and 2) quadratic growth in computational complexity with increasing scales. These feature sequences are then input into the gating Mamba, leveraging its robust capability to fuse contextual information to deeply explore the intrinsic connections and relational patterns within the hierarchical spatial information of the sequence data, thereby achieving a richer feature representation. For the gating Mamba, we introduce a distance-oriented dynamic discriminative gate to replace the traditional binarization method, as shown in Fig. 4. Traditional methods usually adopt a binary static gating approach with a fixed threshold or simple linear transformation, which not only loses a large amount of fine-grained information but also lacks spatial perception ability, especially when dealing with data such as hyperspectral images that have complex spatial hierarchical structures. Dynamic gating calculates spatial distance weights based on sequence centers and uses these weights as learnable parameters to achieve continuous assessment and dynamic adjustment of feature importance, thereby enabling more precise differ-

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entiation and retention of effective features. Meanwhile, the position prior information is effectively retained through the distance weight matrix. Combined with the multidirectional scanning strategy, an implicit spatial encoding is formed, which can perform differentiated processing for the local features of different spatial positions. This weight initialization method based on the distance of the central pixel enhances the model's perception of spatial topological relationships, making the gating process not only rely on feature strength but also combine spatial context information. Furthermore, our OAA enhances the spatial perception capability of the input data by introducing sampling offsets to achieve adaptive sampling of spatial features. This mechanism allows the model to flexibly adjust the regions of interest within a broader receptive field, effectively compensating for the lack of spatial structure information in Mamba's sequential modeling. We provide a detailed introduction to the OAA in Section III-C. In this manner, the model can more effectively parse complex structures and information within the HSI, significantly improving classification performance.

Specifically, given the input feature $F \in R^{B*C*P*P}$, where B denotes the batch size, while C and P represent the spectral channels and spatial dimensions of the input features, respectively. Multiscale convolution is first utilized to comprehensively capture the features of ground objects from different views, ensuring that the model can adapt to diverse ground object morphologies

$$x_{\text{spa}}^{i} = \begin{cases} \text{ReLU}\left(\text{Conv}_{1\times1}(F)\right) \\ \text{ReLU}\left(\text{Conv}_{3\times3}(F)\right) \\ \ldots. \end{cases}, i \in (1,3,\ldots,) \tag{1}$$

$$x_{\text{seq}}^{\text{spa}} = C\text{onv}\left(\text{Concat}\left(x_{\text{spa}}^{i}\right)\right), i \in (1,3,\ldots,). \tag{2}$$

Multidirectional scanning is employed to generate different feature sequences using permutations and combinations. The multidirectional scanning process can be represented as follows:

$$x_h = \text{Reshape}\left(x_{\text{seq}}^{\text{spa}}, (B,N,C)\right) \tag{3}$$

$$x_v = \text{Reshape}\left(\left(x_{\text{seq}}^{\text{spa}}\right)^{\text{T}}, (B,N,C)\right) \tag{4}$$

$$x_{\text{scan}}^{\text{spa}} = \begin{bmatrix} x_h \\ x_v \\ \text{flip}(x_h) \\ \text{flip}(x_v) \end{bmatrix} \tag{5}$$

where N is the length of the spatial feature sequence. Using dynamic projection and selective scanning mechanisms, the representation of the features can be flexibly adjusted to adapt to different input feature structures. The integration of multidirectional features combines the information from various scanning directions to form a richer contextual representation. The resulting feature sequences from the scanning were then input into the gating Mamba as follows:

$$h'(t) = A \cdot h(t) + B \cdot x_{\text{scan}}^{\text{spa}}(t) \tag{6}$$

$$F_{\text{spa}}(t) = C \cdot h(t) + D \cdot x_{\text{scan}}^{\text{spa}}(t). \tag{7}$$

Using this design, Mamba effectively aggregates multiscale features, reduces information redundancy, and enhances multiscale feature representations. For the feature of Mamba, the Euclidean distance between each vector and the reference vector is first calculated using the sequence center vector as the reference. The calculation process is as follows:

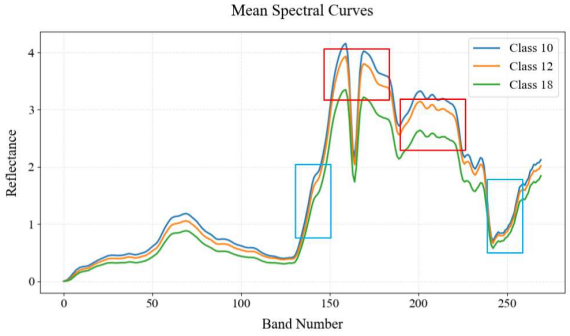


Fig. 5. Characteristic spectral signatures of typical categories.

$$v_{\text{ref}} = F_{\text{spa}}[:, \text{center}, :], \text{center} = (N+1)/2 \tag{8}$$

$$d_{\text{spa}}^{i,j} = \|F_{\text{spa}}^{i,j,:} - v_{\text{ref}}^{i,:}\|, i \in (1,\ldots,B), j \in (1,\ldots,N). \tag{9}$$

Using the mean of these distances as the scale parameter σ

$$\sigma_i = \frac{1}{N}\sum_{j=1}^{N} d_{\text{spa}}^{i,j} \tag{10}$$

$$w_{i,j} = \exp\left(-0.5 \times \left(\frac{d_{\text{spa}}^{i,j}}{\sigma_i}\right)^2\right), w_{i,j} \in R^{B*N*C}. \tag{11}$$

The returned normalized weights reflect the relative importance of each vector in the sequence with respect to the reference vector. The result is used as the initial weight for the dynamic discrimination gate and added to the spatial features in the form of a Hadamard product. Finally, the output features of the HGM are obtained as follows:

$$F_{\text{spa}} = (1+w) * F_{\text{spa}}(t) + \text{OAA}\left(x_{\text{seq}}^{\text{spa}}\right), F_{\text{spa}} \in R^{B*N*C} \tag{12}$$

where OAA(·) represents the feature processing of the OAA mechanism.

B. Random Shuffled Gating Mamba

In hyperspectral data, different categories exhibit significant differences in reflectance characteristics within specific spectral bands. For HongHu dataset, the average spectral response curves of three typical categories are shown in Fig. 5. The spectral bands marked by the red box demonstrate pronounced interclass separability, providing critical spectral features for identification. In contrast, within the blue box region, the spectral curves exhibit substantial overlap, indicating notable spectral confusion. Additionally, band combinations play a pivotal role in interclass separability. A comparison between (a) conventional band ordering and (b) shuffled band ordering in Fig. 6 reveals that the latter enhances color differences and boundary clarity between categories, suggesting that altering the band combination sequence can uncover additional spectral information beneficial for classification. Given the significant role of nonlinear relationships in category differentiation, we

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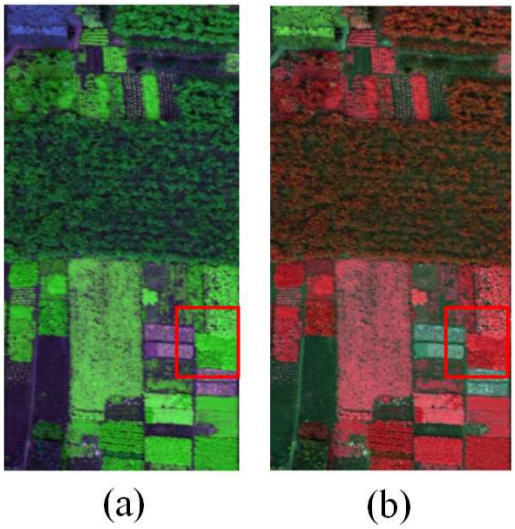


Fig. 6. Color composite comparison. (a) Wavelength-ordered versus (b) shuffled-band configuration.

propose an innovative spectral scanning strategy aimed at enhancing the efficacy of feature extraction by introducing two novel scanning paths, namely the "shuffled path." This path disrupts the fixed order of the original spectral sequence. Randomly shuffling the spectral data and then rescanning allows the model to capture spectral features with higher discriminative power that are overlooked or weakened in traditional sequential scanning. Furthermore, by integrating the shuffled path with other directional scanning paths, the model forms a multidimensional, multilevel spectral feature extraction network, constructing a more comprehensive and dynamic spectral analysis strategy. This strategy explores a broader and more flexible combination of features within the spectral feature space, encouraging the model to discover nonintuitive and nonlinearly correlated feature combinations, thereby achieving high-precision recognition and classification of different categories. Second, this random shuffling strategy essentially constitutes a targeted feature enhancement mechanism. Its design philosophy parallels techniques like random cropping in computer vision or word order perturbation in NLP—all introducing controlled variations to increase training data diversity. Each shuffling operation can be viewed as a random sampling of spectral information, effectively enhancing the model's robustness in spectral sequence modeling. Building on this scanning strategy, we constructed the RSGM. After the shuffled spectral sequences are input into the gating Mamba, the model adaptively adjusts its internal parameters based on the current input and previous states, thereby capturing and integrating long-term dependencies within the spectral sequences. The dynamic gating mechanism incorporates band position priors in the spectral dimension through a learnable interband distance weight matrix, enabling physical structure awareness and adaptive selection of spectral features. This approach simultaneously considers both the independent feature contributions of individual bands and the physical distribution characteristics of spectral sequences. Consequently, the gating process not only adheres to the physical principles governing spectral data but also adaptively learns optimal feature combinations, effectively suppressing noise and redundant bands while significantly enhancing the capture capability for subtle spectral features. Furthermore, we extend the omni-dimensional adaptive attention (OAA) to the spectral dimension to identify critical band information and capture nonlocal spectral dependencies.

Specifically, as shown in Fig. 7, given the input feature $F \in R^{B\times C\times P\times P}$, depthwise separable convolution is performed along the spectral dimension for positional modeling. This strategy aims to precisely capture the implicit local features and spectral structures hidden in the spectrum

$$x_{\text{seq}}^{\text{spe}} = \text{PatchEmbed}\,(F)\,, x_{\text{seq}}^{\text{spe}} \in R^{B*M*N} \tag{13}$$

where M is the length of the spectral feature sequence.

Subsequently, by integrating shuffle scanning with a multilayer perceptron (MLP), a deep transformation of the spectral data is performed, which is then mapped into structured sequence data to obtain multipath scanning sequences. These sequences encapsulate rich information on the spectrum across various dimensions and perspectives. The random shuffle scanning process can be represented as follows:

$$x_{\text{hwwh}} = x_{\text{seq}}^{\text{spe}} \tag{14}$$

$$x_{\text{reverse}} = \text{flip}\left(x_{\text{seq}}^{\text{spe}}, \text{dims} = [-1]\right) \tag{15}$$

$$x_{\text{shuffle}} = x\left[:, :, \text{randperm}\,(M)\right] \tag{16}$$

$$x_{\text{reverse_}s} = \text{flip}\left(x_{\text{reverse}}\left[:, :, \text{randperm}\,(M)\right], \text{dims} = [-1]\right) \tag{17}$$

$$x_{\text{scan}}^{\text{spe}} = [x_{\text{hwwh}}, x_{\text{reverse}}, x_{\text{shuffle}}, x_{\text{reverse_}s}]\,. \tag{18}$$

The information within these sequences is input into the dynamic gating Mamba, capturing and integrating the long-term dependencies and complex features within the spectral sequences

$$y_{\text{out}} = \text{selective_scan}\left(x_{\text{scan}}^{\text{spe}}\right)$$

$$F'_{\text{spe}} = w_1 y_{\text{hwwh}} + w_2 y_{\text{reverse}} + w_3 y_{\text{shuffle}} \tag{19}$$

$$+ w_4 y_{\text{reverse_}s}, F'_{\text{spe}} \in^{B*M*N} \tag{20}$$

where $w_i, i = 1, 2, 3, 4$ represent the weights of the four-sequence data, respectively.

Similar to the HGM, in the gating Mamba, distance-oriented spectral dynamic discrimination gate is applied to extract the effective features. The difference lies in the use of the spectral sequence index to calculate the relative distances, as shown in Fig. 8. First, using the middle index as a reference, we have

$$I_{\text{ref}} = F'_{\text{spe}}\,[:, \text{center}, :]\,, \text{center} = (M+1)/2 \tag{21}$$

$$d_{\text{spe}}^{i,j} = \|F_{\text{spe}}^{i,j} - I_{\text{ref}}\|,\ i \in (1, \ldots, B)\,,\ j \in (1, \ldots, M)\,. \tag{22}$$

Thus, the initial weights of the dynamic discrimination gate in the spectral branch are obtained

$$\alpha \in R^{B*M}. \tag{23}$$

Finally, the output features of the spectral branch are obtained as follows:

$$F_{\text{spe}} = (1+\alpha) * F'_{\text{spe}} + \text{OAA}\left(x_{\text{seq}}^{\text{spe}}\right), F_{\text{spe}} \in^{B*M*N}. \tag{24}$$

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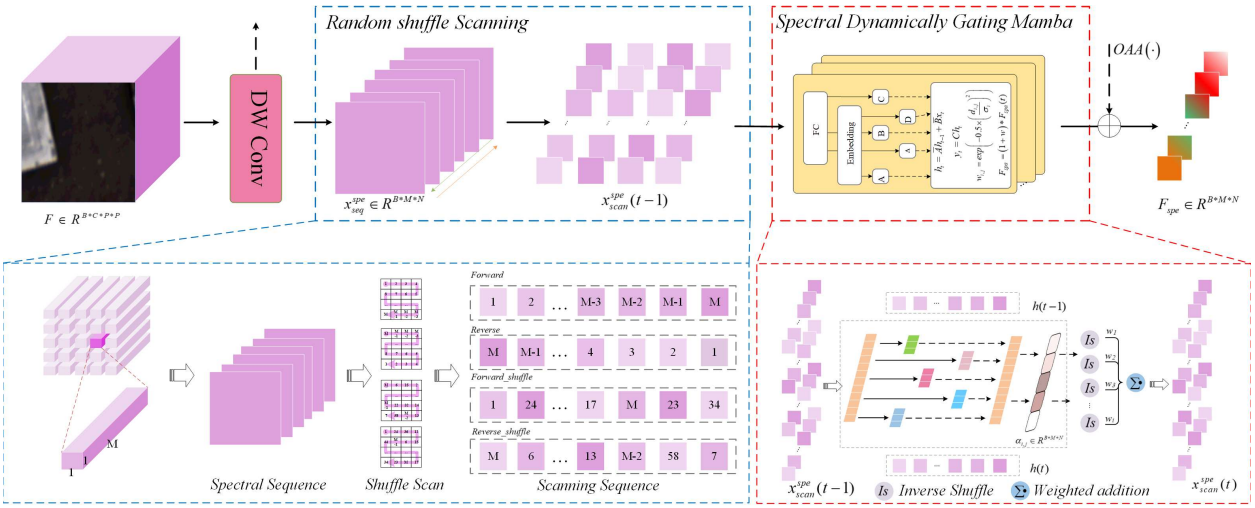


Fig. 7. Overall structure of RSGM, which includes random shuffle scanning and spectral dynamic gating Mamba. Here, $F \in R^{B*C*P*P}$ and $F_{\text{spe}} \in R^{B*M*N}$ represent the input and output features of RSGM, respectively, $x^{\text{spe}}_{\text{scan}}(t-1)$ and $h(t-1)$ denote the input sequence and state variables of the Gate Mamba, respectively, and $\alpha_{i,j} \in R^{B*M*N}$ represents the dynamic gating weights.

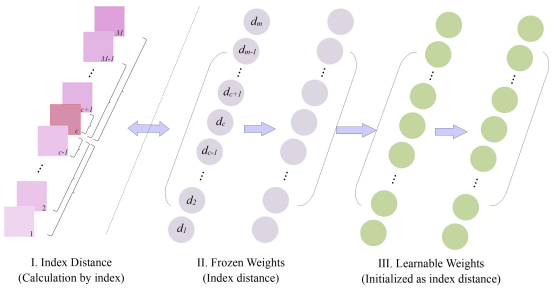


Fig. 8. Learning process of dynamic gating weights in RSGM. (a) Stage I: index-distance computation. (b) Stage II: frozen gate weights (from index distance). (c) Stage III: learnable gate weights (initialized by index distance).

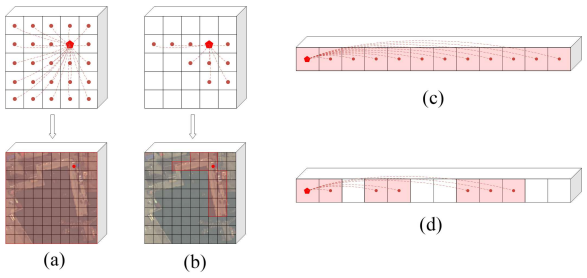


Fig. 9. Comparison of OAA with the traditional attention mechanisms. (a)–(d) OAA sampling scenarios in the spatial and spectral branches, respectively. The red pentagons represent queries and the red dots indicate sampling locations.

C. Offset-Aware Attention

Traditional attention mechanisms typically sample at fixed spatial locations and compute the global elements, resulting in excessive memory and computational costs. The captured features are also more susceptible to irrelevant parts beyond the region of interest. To address this, we designed an OAA by combining multihead attention with spatial sampling techniques. OAA introduces sampling offsets, enabling the attention mechanism to flexibly sample objects at different

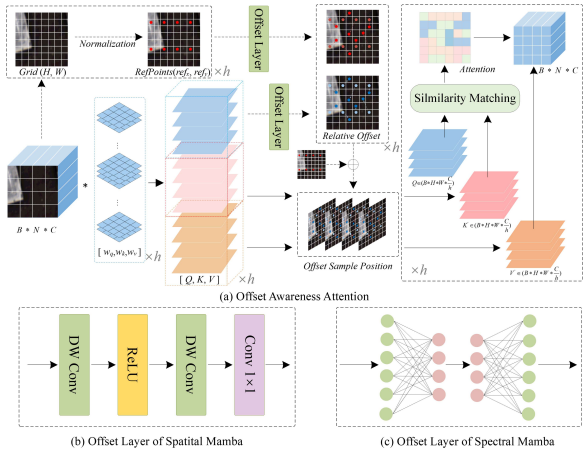


Fig. 10. Overall process of OAA. (a) General structure of OAA and (b) and (c) structures of the offset layer in the spatial and spectral branches, respectively. Here, $F \in R^{B*N*C}$ and $F_{\text{att}} \in R^{B*N*C}$ represent the input and output features of OAA.

spatial locations. This approach allows the model to capture spatial features over a broader range, enhancing its spatial perception capability of input data, and thereby improving its ability to handle complex spatial structures. This capability is particularly suitable for supplementing the spatial structural information missing from Mamba sequence data. In addition, by calculating the sampling offsets and performing bilinear interpolation sampling, the model can dynamically adjust the spatial locations on which it focuses. This dynamic variation enables the OAA to concentrate more on the relevant regions and capture more information features from the corresponding important areas, as illustrated in Fig. 9.

Specifically, as shown in Fig. 10(a), normalized grid coordinates are first generated based on the spatial shape of the feature map, providing a standardized method to represent the position of each pixel in the feature map. These coordinates can be adapted for inputs of different sizes to ensure sampling

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consistency. Given the height and width of the feature map as HH and WW, respectively, the normalized grid coordinates are

$$\mathrm{ref}_y = \frac{1}{H}[0.5, 1.5, \ldots, H-0.5] \tag{25}$$

$$\mathrm{ref}_x = \frac{1}{W}[0.5, 1.5, \ldots, W-0.5] \tag{26}$$

where ref_x and ref_y are the coordinate values. By gridding ref_x and ref_y, we obtain

$$\mathrm{ref}_{\mathrm{grid}} = \left\{ \left(\frac{i+0.5}{H}, \frac{j+0.5}{W} \right) | i = 0, 1, \ldots, H-1; \; j = 0, 1, \ldots, W-1 \right\}. \tag{27}$$

For compatibility with Mamba's sequence data, we set the input features of the OAA as $F \in R^{B*N*C}$. For the input features of each head, linear transformation is performed to generate the queries, keys, and values, where $Q_i = w_i^q F$, $K_i = w_i^k F$, and $V_i = w_i^v F$, where w_i^q, w_i^k, and w_i^v are the weight matrices for the linear transformations and $i \in (1, 2, \ldots, h)$. h is the number of attentional heads. Subsequently, using the offset layer, the reference points and query features are mapped to the offset space, obtaining the offset for each reference point as follows: For the spectral and spatial features, different offset layer structures are adopted, as shown in Fig. 10(b) and (c)

$$\Delta_i = \mathrm{Max}\left(w_i^{\Delta} Q_i, w_i^{\Delta} \mathrm{ref}\right) \tag{28}$$

where w_i^{Δ} is the weight of the offset layer. By using the reference points and offsets to calculate the offset positions, for each reference point $\mathrm{re}f_{\mathrm{grid}}(m, n)$, its offset position is

$$\mathrm{offset}(m, n) = \mathrm{re}f_{\mathrm{grid}}(m, n) + \Delta(m, n). \tag{29}$$

Finally, the offset positions are used to perform bilinear interpolation sampling from the K and V, obtaining the new K' and V'

$$K'(p, q) = \sum_{m,n} w_{m,n}(\mathrm{offset}(p, q)) \cdot K(m, n) \tag{30}$$

$$V'(p, q) = \sum_{m,n} w_{m,n}(\mathrm{offset}(p, q)) \cdot V(m, n) \tag{31}$$

where $w_{i,j}$ is the weight for bilinear interpolation, which depends on the offset position $\mathrm{offset}(p, q)$ and the positions and distances of the four neighboring pixel points. The attention output is computed separately for each head. The outputs of all the heads are concatenated as follows:

$$F_{\mathrm{att}} = \sum_h \mathrm{softma}\left(\frac{Q_i \cdot (K_i')^{\mathrm{T}}}{\sqrt{d_k}} \right) \cdot V_i', d_k = C/h$$
$$F_{\mathrm{att}} \in R^{B*N*C}. \tag{32}$$

Through multihead and spatial sampling mechanisms, the model can compute attention in parallel across multiple subspaces by capturing different feature patterns and spatial relationships. This design enhanced the expressive power and adaptability of the model to complex spatial structures. Notably, our OAA can achieve a similar process in the spectral dimension by simply adjusting the data dimensions and applying an offset. This approach allows the model to flexibly sample across different spectral channels, thereby better capturing the spectral features and correlations.

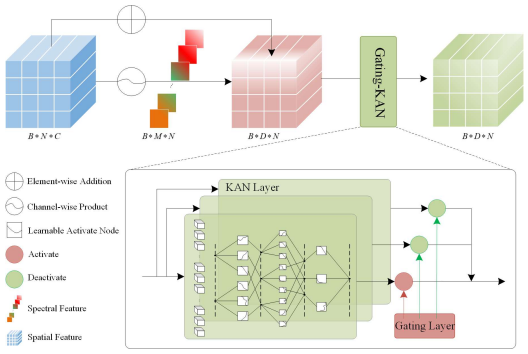


Fig. 11. Overall structure of SOFRM, where $F \in R^{B*D*N}$ represents the output feature of SOFRM.

D. Spectral-Oriented Feature Review Module

After capturing the spatial details and spectral characteristics of an HSI through two independent scanning branches, designing an efficient and adaptable structure to fuse the information from these two dimensions becomes crucial. Different datasets exhibit significant diversity in their reliance on spatial and spectral information, owing to variations in their intrinsic characteristics. For instance, in the Qingyun dataset, spectral information is particularly crucial because of the widespread shadowed areas, which play an irreplaceable role in distinguishing the ground cover within these regions. In contrast, the Tangdaowan and Pingan datasets, with their complex geographical structures and numerous small target objects, rely more heavily on spatial information, making spatial details a key factor for accurate identification and classification. Traditional information fusion methods, such as simple addition or stacking, often overlook these differential needs among datasets while achieving information merging to some extent, leading to information redundancy and the submergence of critical features, thereby affecting the accuracy of subsequent analysis and classification. To overcome this limitation, we propose SOFRM. This module treats spectral information as a dynamic balancing factor and enhances spatial features based on variations in spectral characteristics. This mechanism not only ensures effective information integration, but also flexibly adapts to the different emphases on spatial and spectral information across datasets, thereby highlighting more discriminative features. Furthermore, to enhance the expressive power and robustness of the fused features, we introduced the Kolmogorov–Arnold representation theorem and designed a gating-KAN for feature enhancement. Gating-KAN not only possesses the ability to learn new features, but more importantly, can significantly optimize already learned features, achieving a deep review and fine-tuning of features.

Specifically, as shown in Fig. 11, given the spectral and spatial features extracted by the Mamba branches as $F_{\mathrm{spe}} \in R^{B\times M\times N}$ and $F_{\mathrm{spa}} \in R^{B\times N\times C}$, respectively, the features are first compressed and reconstructed into the following designed dimensions:

$$F'_{\mathrm{spe}} = w_{\mathrm{spe}} F_{\mathrm{spe}} + b_{\mathrm{spe}}, F'_{\mathrm{spe}} \in R^{B\times D} \tag{33}$$

$$F'_{\mathrm{spa}} = \sum_i \sum_j w_{i,j} \cdot F_{\mathrm{spa}_{i,j}}, F'_{\mathrm{spa}} \in R^{B\times D\times N} \tag{34}$$

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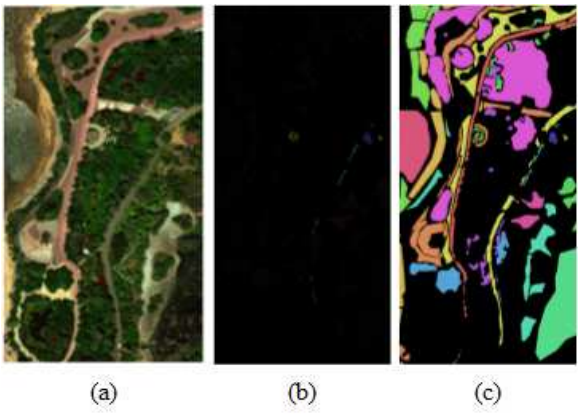


Fig. 12. QUH-Tangdaowan dataset. (a) Annotation of the training, (b) validation, and (c) test samples.

where w_{spe} and b_{spe} are the parameters and biases in feature reconstruction, respectively, and D is the preset dimension. Next, the spectral information is fused with the spatial information as a balancing factor

$$F_{\text{fus}} = \left(1 + F'_{\text{spe}}\right) * F'_{\text{spa}}, F_{\text{fus}} \in R^{B\times D\times N}. \tag{35}$$

Next, the fused features are input into the gating-KAN network in the form of tokens. The KAN places learnable activation functions on the edges using 1-D functions parameterized as spline curves to serve as parameters for each weight. This approach simultaneously applies nonlinear and linear transformations to the features, thereby enhancing the representation capability and classification performance of the model for complex data. Subsequently, a discrimination gate is used to filter the enhanced information and obtain effective features

$$F_{\text{en}} = \sum_{q=1}^{Q} \phi_q \left(\sum_{p=1}^{P} w_{\text{pq}} \cdot F_{\text{fus}_p} + b_q \right) \tag{36}$$

$$\phi = \left\{\varphi_{i,j}\right\}, \ i \in (1, 2, \ldots, P), \ j \in (1, 2, \ldots, Q) \tag{37}$$

$$F_{\text{out}} = \text{Gate}\left(\theta, F_{\text{en}}\right), F_{\text{out}} \in R^{B\times D\times N} \tag{38}$$

where F_{en} is the features after enhancement and review, F_{out} is the final output feature, ϕ is a 1-D nonlinear activation function, and Q and P are the dimensions of the output and input, respectively.

IV. Experimental

This section provides an overview of the experimental data, evaluation metrics, and setup. Finally, a comprehensive comparison of various methods used in the experiments is presented.

A. Datasets and Experimental Setting

1) Datasets: The QUH-Tangdaowan dataset [58] was acquired on May 18, 2021, at the Tangdaowan National Wetland Park in Qingdao, China. The UAV flowed at an altitude of 300 m, yielding a spatial resolution of approximately 0.15 m. The dataset comprises 176 spectral bands covering a wavelength range of 400–1000 nm with an image size of 1740 × 860 pixels. All annotated pixels were categorized into 18 classes. This dataset presents significant challenges for accurate classification owing to its diverse irregular land-cover types and highly similar spectral curves. For clarity, these classes are denoted as C1–C18 in the subsequent figures and tables. Fig. 12 illustrates a visual representation of the dataset, and Table I details the training configuration.

TABLE I
Number of Training, Validation, and Test Ground Truth Data in QUH-Tangdaowan Dataset

No.	Class	Train	Test	Total
1	Rubber track	500	25349	25849
2	Flaggingv	500	55053	55553
3	Sandy	500	33537	34037
4	Asphalt	500	60190	60690
5	Boardwalk	500	1362	1862
6	Rocky shallows	500	36625	37125
7	Grassland	500	13627	14127
8	Bulrush	500	63587	64087
9	Gravel road	500	30195	30695
10	Ligustrum vicaryi	500	1283	1783
11	Coniferous pine	500	20736	21236
12	Spiraea	200	549	749
13	Bare soil	500	1186	1686
14	Buxus sinica	200	686	886
15	Photinia serrulata	500	13520	14020
16	Populus	500	140404	140904
17	Ulmus pumila L	500	9302	9802
18	Seawater	500	41775	42275
-	Total	8400	548966	557366

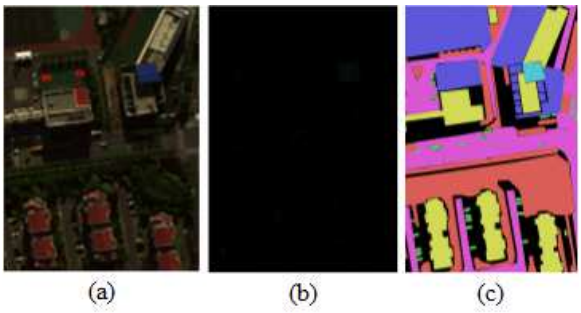


Fig. 13. QUH-Qingyun dataset. (a) Annotation of the training, (b) validation, and (c) test samples.

The QUH-Qingyun dataset [58] was acquired on May 18, 2021, from Qingyun Road Elementary School and its surrounding residential areas in Qingdao, China. The image dimensions were 880 × 1360 pixels, comprising 270 spectral bands with a wavelength range of 400–1000 nm and a spatial resolution of 0.15 m. All annotated pixels were categorized into six classes. The dataset includes areas obscured by building shadows and encompasses categories such as trees, cars, and asphalt roads, where shadow interference significantly increases recognition difficulty. Fig. 13 illustrates a visual representation of the dataset, and Table II lists the pixel usage for each class.

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TABLE II
PIXELS USED FOR TRAINING AND TESTING FROM THE QUH-QINGYUN DATASET

No.	Class	Train	Test	Total
1	Trees	500	277650	278150
2	Concrete building	500	179012	179512
3	Car	500	13283	13783
4	Ironhide building	500	9267	9767
5	Plastic playground	500	217235	217735
6	Asphalt road	500	255446	255946
-	Total	3000	951893	954893

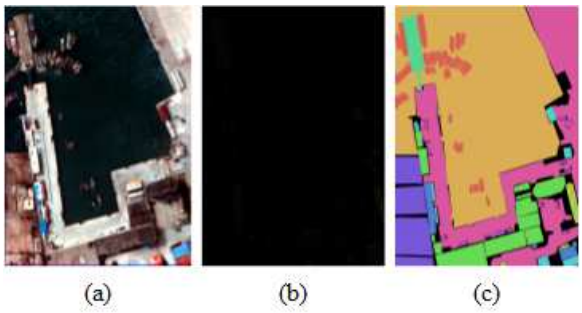


Fig. 14. QUH-Pingan dataset. (a) Annotation of the training, (b) validation, and (c) test samples.

TABLE III
PIXELS USED FOR TRAINING AND TESTING FROM THE QUH-PINGAN DATASET

No.	Class	Train	Test	Total
1	Ship	500	48435	48935
2	Seawater	500	577613	578113
3	Trees	500	7845	8345
4	Concrete structure building	500	88473	88973
5	Floating pier	500	20259	20759
6	Brick houses	500	13586	14086
7	Steel houses	500	13491	13991
8	Wharf construction land	500	82613	83113
9	Car	500	7608	8108
10	Road	500	276014	276514
-	Total	5000	1135937	1140937

The QUH-Pingan dataset [58] was captured via UAV at Qingdao's Huangdao ferry terminal on May 19, 2021, featuring 0.10 m resolution hyperspectral imagery (1230 × 1000 pixels) with 176 spectral bands (400–1000 nm) across ten land-cover classes. Its combination of high spatial resolution and significant target size variations makes it particularly valuable for evaluating multiscale classification approaches, with Fig. 14 and Table III providing visual and quantitative references, respectively.

The WHU-Hi-HongHu dataset [59] was acquired via UAV over agricultural fields in Honghu City, China, at 100 m altitude, producing 0.043 m resolution imagery (940 × 475 pixels) with 270 spectral bands (400–1000 nm). This dataset captures the spectral characteristics of fragmented farmland containing different cultivars of the same crop species, making it particularly valuable for fine-grained agricultural classification studies, with Fig. 15 and Table IV providing visual and quantitative references, respectively.

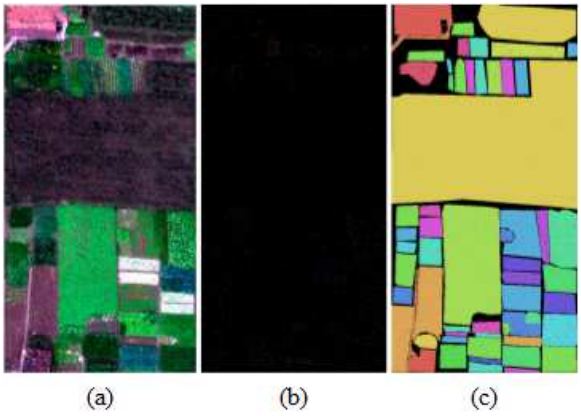


Fig. 15. WHU-Hi-HongHu dataset. (a) Annotation of the training, (b) validation, and (c) test samples.

TABLE IV
PIXELS USED FOR TRAINING AND TESTING FROM THE WHU-HI-HONGHU DATASET

No.	Class	Train	Test	Total
1	Red roof	100	13941	14041
2	Road	100	3412	3512
3	Bare soil	100	21721	21821
4	Cotton	100	163185	163285
5	Cotton firewood	100	6118	6218
6	Rape	100	44457	44557
7	Chinese cabbage	100	24003	24103
8	Pakchoi	100	3954	4054
9	Cabbage	100	10719	10819
10	Tuber mustard	100	12294	12394
11	Brassica parachinensis	100	10915	11015
12	Brassica chinensis	100	8854	8954
13	Small Brassica chinensis	100	22407	22507
14	Lactuca sativa	100	7256	7356
15	Celtuce	100	902	1002
16	Film covered lettuce	100	7162	7262
17	Romaine lettuce	100	2910	3010
18	Carrot	100	3117	3217
19	White radish	100	8612	8712
20	Garlic sprout	100	3386	3486
21	Broad bean	100	1228	1328
22	Tree	100	3940	4040
-	Total	2200	384493	386693

2) Experimental Settings: To validate the superiority of the proposed method, it was compared with several state-of-the-art models including CVT [39], SpectralFormer [40], SSFTT [41], BS2T [46], MambaHSI [25], VIM [53], MiM [57], 1-D_KAN [60], 2-D_KAN [60], and 3-D_KAN [60]. This comparison is based on the same number of training samples.

In terms of parameter configuration, all models employed the AdamW optimizer with a batch size of 64 and were trained for 50 epochs. Specifically, CVT, SpectralFormer,

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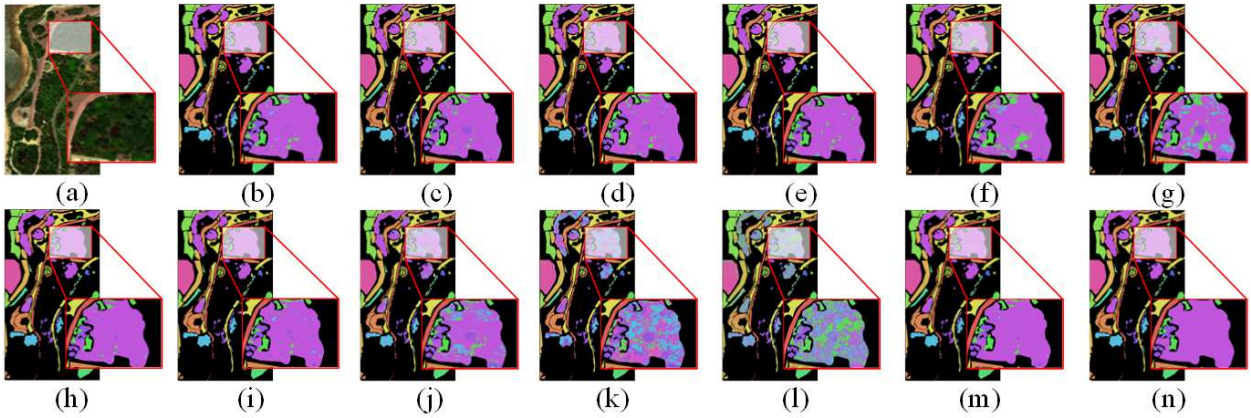


Fig. 16. Visual results of various methods on the QUH-Tangdaowan dataset. The first row shows the classification results, and the second is an enlargement of the red box in the first row. (a) HSI. (b) CVT. (c) SpectralFormer. (d) SSFTT. (e) BS2T. (f) MambaHSI. (g) VIM. (h) MiM. (i) GraphMamba. (j) 1-D_KAN. (k) 2-D_KAN. (l) 3-D_KAN. (m) SEDGM. (n) Gt.

TABLE V
PERFORMANCE METRICS OF VARIOUS METHODS ON THE QUH-TANGDAOWAN DATASET WITH OPTIMAL RESULTS IN BOLD

NO.	CVT	SpectralFormer	SSFTT	BS2T	MambaHSI	VIM	MiM	GraphMamba	1D_KAN	2D_KAN	3D_KAN	SEDGM
1	99.86	99.21	99.81	99.74	99.88	99.57	99.81	99.62	99.80	99.22	**99.89**	99.71
2	98.79	94.82	98.95	96.17	96.76	94.42	99.21	98.86	97.40	86.34	93.68	**99.65**
3	94.99	83.44	96.26	97.21	92.39	83.75	97.48	97.02	89.90	79.49	**97.92**	97.19
4	99.64	96.95	99.82	99.98	98.68	96.04	99.80	96.03	99.74	95.97	89.38	**99.84**
5	98.70	88.66	91.47	94.78	60.71	63.63	97.99	97.94	85.65	37.09	9.31	**99.78**
6	**98.15**	86.12	96.96	94.76	95.26	94.74	96.40	94.15	95.43	89.28	82.07	97.37
7	78.75	69.66	70.75	67.07	54.08	48.07	80.99	93.56	62.59	44.01	19.08	**86.23**
8	99.90	98.42	99.83	99.83	99.97	99.84	**99.96**	99.00	99.25	97.62	93.84	99.70
9	98.19	97.47	99.67	99.63	97.39	92.79	99.18	99.10	98.10	87.04	87.93	**99.85**
10	98.31	84.38	96.18	98.62	99.53	87.30	99.69	**99.92**	97.34	89.95	43.86	97.94
11	92.15	49.03	91.08	96.16	87.21	72.77	92.72	**96.79**	74.56	34.92	29.21	95.18
12	57.73	18.02	35.63	95.25	66.63	68.99	55.45	**99.60**	41.59	48.84	3.660	58.72
13	99.58	97.95	98.26	100.0	99.83	96.42	97.94	100.0	99.83	88.84	47.95	**100.0**
14	70.43	63.44	57.70	68.12	54.93	40.93	87.17	**98.70**	44.57	32.66	30.71	83.97
15	87.86	75.75	85.74	91.15	84.06	74.71	84.01	**92.59**	80.68	68.51	87.26	89.49
16	99.77	71.27	99.36	99.36	98.66	97.90	99.65	89.58	98.61	90.28	91.51	**99.88**
17	95.43	75.98	81.38	90.15	87.97	90.58	96.49	96.10	79.94	59.34	43.57	**97.13**
18	99.09	98.82	99.93	99.92	99.81	96.40	100.0	99.97	99.41	99.61	95.01	**100.0**
OA	95.96	84.22	95.11	95.37	93.13	89.78	96.17	95.49	92.66	80.78	71.31	**96.80**
AA	97.69	92.38	97.37	97.20	96.02	93.76	97.80	97.14	96.03	89.31	81.16	**98.11**
Kappa	95.42	82.45	94.47	94.81	92.25	88.51	95.67	94.88	91.73	78.65	68.80	**96.37**

SSFTT, BS2T, and VIM uniformly used an initial patch size of 11 × 11. MambaHSI was configured with 256 feature dimensions, 128 output dimensions, Δ initialized to 0.01, and a learning rate of 0.0005. VIM adopted 128-D features, a three-layer network depth, and a learning rate of 0.0001. MiM standardized feature dimensions to 64 with a T-Mamba encoder depth of 2. The KAN hidden layer utilized 64-D features, eight grid points, and a learning rate of 0.001. Our SEDGM implemented 128-D features, a learning rate of 0.0005, and eight bias sampling points, with an initial patch size of 9 × 9 for the HongHu dataset and 11 × 11 for others. Regarding the evaluation metrics, we selected the overall accuracy (OA), average accuracy (AA), kappa coefficient, and per-class accuracy, which provided a comprehensive view of the classification performance. All experiments were conducted on an Ubuntu system using PyTorch 2.0 and a GeForce RTX 4070Ti GPU.

B. Classification Performance

1) Results for QUH-Tangdaowan Dataset: The QUH-Tangdaowan dataset exhibited an irregular land-cover distribution accompanied by significant spectral confusion between classes, presenting substantial challenges for classification. Table V presents the quantitative results of different methods for this dataset. Transformer-based models and their variants achieved favorable classification performance owing to their powerful long-range modeling capabilities. Notably, CVT and BS2T achieved overall accuracies exceeding 95% and kappa coefficients exceeding 94%. Furthermore, the Mamba models

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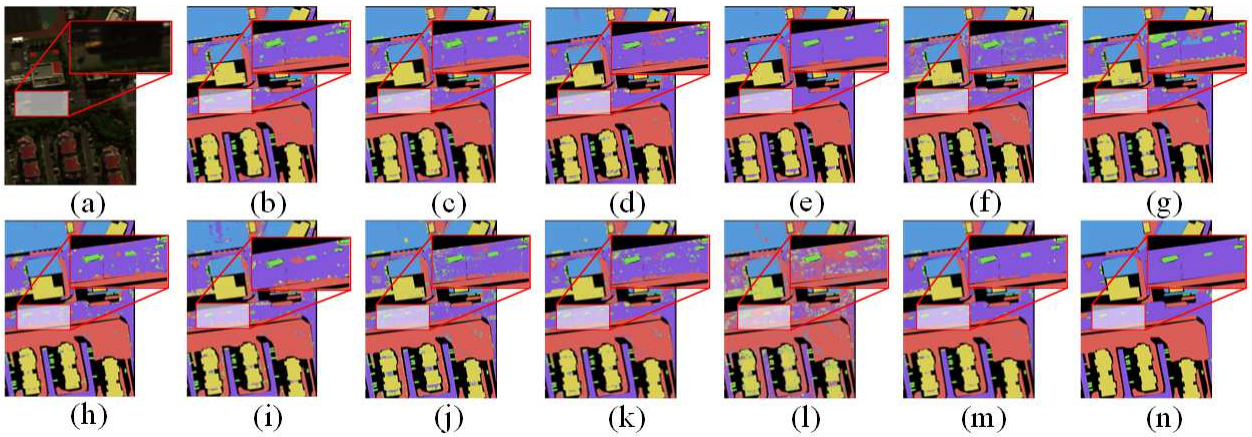


Fig. 17. Visual results of various methods on the QUH-Qingyun dataset. The first row shows the classification results, and the second is an enlargement of the red box in the first row. (a) HSI. (b) CVT. (c) SpectralFormer. (d) SSFTT. (e) BS2T. (f) MambaHSI. (g) VIM. (h) MiM. (i) GraphMamba. (j) 1-D_KAN. (k) 2-D_KAN. (l) 3-D_KAN. (m) SEDGM. (n) Gt.

TABLE VI
PERFORMANCE METRICS OF VARIOUS METHODS ON THE QUH-QINGYUN DATASET WITH OPTIMAL RESULTS IN BOLD

NO.	CVT	SpectralFormer	SSFTT	BS2T	MambaHSI	VIM	MiM	GraphMamba	1D_KAN	2D_KAN	3D_KAN	SEDGM
1	91.28	91.14	91.48	94.79	93.43	92.31	94.21	92.27	92.12	90.67	77.25	**95.08**
2	94.50	92.47	94.49	95.67	87.74	92.58	95.33	90.95	92.94	91.77	77.69	**96.21**
3	50.02	40.15	53.07	75.15	32.68	26.86	43.16	**90.28**	30.51	32.27	38.81	61.78
4	88.99	89.90’	96.66	92.15	94.24	92.23	89.57	**99.77**	97.95	90.28	93.89	90.46
5	95.61	97.59	97.59	97.02	90.88	93.19	97.23	92.01	95.42	95.37	88.15	**98.54**
6	**96.71**	94.79	90.86	89.16	89.71	92.13	93.81	85.23	90.49	92.42	69.98	94.98
OA	91.19	90.17	90.52	91.89	86.86	88.20	91.58	90.12	88.36	88.47	75.49	**93.30**
AA	93.32	92.38	91.67	89.15	88.98	87.77	93.18	91.75	90.19	89.59	78.61	**94.94**
Kappa	88.46	87.14	87.54	89.32	82.87	84.59	88.98	87.03	84.81	84.95	67.76	**91.21**

demonstrated competitive performance, with MiM achieving an OA of 96.17% and kappa coefficient of 95.67%. In contrast, KAN showed a relatively inferior performance on this dataset, with the highest OA of only 92.66%. Although these models performed well overall, they still faced significant misclassification issues in certain specific categories (e.g., C5-Boardwalk, C7-Grassland, C11-Coniferous pine, C14-Buxus sinica). SEDGM achieved excellent accuracy in these categories, particularly for C5, C7, and C11, and attained the highest OA, average accuracy (AA), and kappa. Fig. 16 visually demonstrates the classification results of the different methods, with an emphasis on enlarged local regions. The magnified details reveal that the other models generally exhibited more misclassifications, particularly for categories such as coniferous pine, *Buxus sinica*, *Photinia serrulata*, and *Populus*. This phenomenon can be attributed to the high spectral similarity and complex distribution patterns of these categories, which significantly increase the difficulty for classifiers to extract discriminative features. GraphMamba demonstrates enhanced classification accuracy for specific species like C14 and C15, through its distinctive graph-structured modeling capability that effectively captures topological relationships among land-cover features. However, its overall classification performance remains suboptimal, suggesting limitations in comprehensive feature representation. In contrast, SEDGM effectively enhanced the extraction of structural features and nonlinear spectral characteristics through the introduction of biased spatially aware attention and shuffled spectral scanning mechanisms, resulting in smoother classification outcomes. Overall, SEDGM demonstrated superior classification performance in scenarios involving complex feature distributions and similar category confusion.

2) Results for QUH-Qingyun Dataset: The primary challenge of the QUH-Qingyun dataset lies in the interference caused by shadow occlusions in classification tasks. The quantitative and qualitative evaluation results are presented in Table VI and Fig. 17. CVT, SSFTT, and BS2T achieved the highest accuracy in classifying asphalt roads, iron-hide buildings, and cars, respectively. SEDGM achieved the best performance for classifying trees, concrete buildings, and iron-hide buildings, with accuracies of 95.08%, 96.21%, and 98.54%, respectively. Moreover, its OA, AA, and kappa coefficient ranked first among all methods. Fig. 17 magnifies the classification results of the shadowed areas for a detailed analysis. The shadowed regions include categories such as trees, asphalt roads, and cars. CVT, MambaHSI, and VIM exhibited significant deficiencies in car recognition, mistakenly classifying large areas of asphalt roads as cars. Meanwhile, SpectralFormer, SSFTT, and 3-D_KAN tended to misclassify shadowed areas as trees. Through an in-depth analysis, we believe that pixels in shadowed areas typically exhibit low and similar grayscale values, which further exacerbates spectral confusion and increases the difficulty of classification tasks. The incorporation of the shuffled spectral scanning

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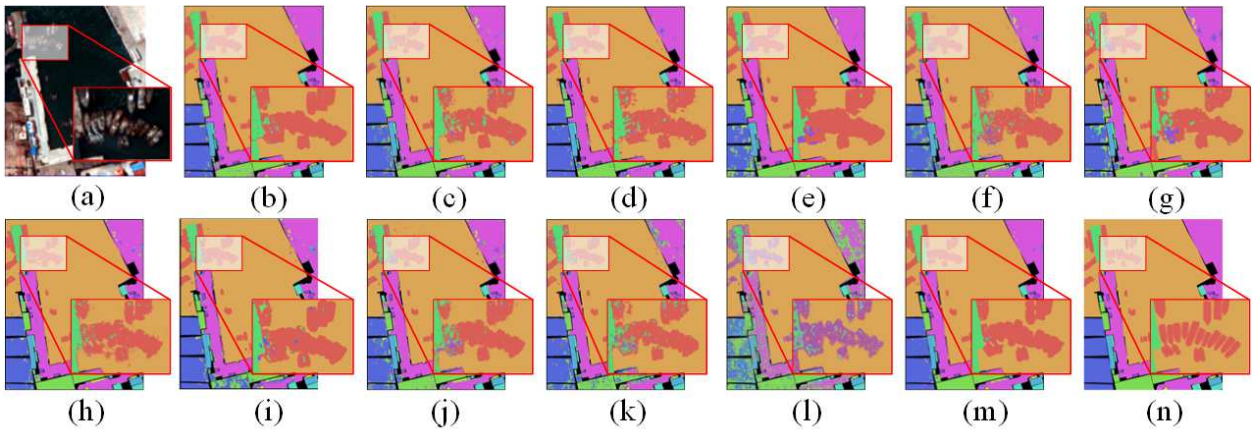


Fig. 18. Visual results of various methods on the QUH-Pingan dataset. The first row shows the classification results, and the second is an enlargement of the red box in the first row. (a) HSI. (b) CVT. (c) SpectralFormer. (d) SSFTT. (e) BS2T. (f) MambaHSI. (g) VIM. (h) MiM. (i) GraphMamba. (j) 1-D_KAN. (k) 2-D_KAN. (l) 3-D_KAN. (m) SEDGM. (n) Gt.

TABLE VII
PERFORMANCE METRICS OF VARIOUS METHODS ON THE QUH-PINGAN DATASET WITH OPTIMAL RESULTS IN BOLD

NO.	CVT	SpectralFormer	SSFTT	BS2T	MambaHSI	VIM	MiM	GraphMamba	1D_KAN	2D_KAN	3D_KAN	SEDGM
1	77.84	76.23	70.95	70.96	74.25	72.41	77.54	**91.77**	75.01	69.51	57.84	84.60
2	99.78	99.67	**99.85**	99.75	99.79	99.35	99.53	97.46	99.70	99.70	98.38	99.80
3	72.25	60.30	90.98	66.89	66.25	58.01	69.86	**99.82**	64.66	80.46	96.27	82.99
4	**96.09**	87.12	95.57	81.15	81.14	77.60	93.64	94.57	81.32	71.03	41.92	95.02
5	83.16	73.96	70.06	86.66	62.01	65.85	83.39	**94.29**	69.56	67.98	33.79	86.49
6	89.63	93.40	94.18	93.37	82.03	77.35	91.52	**98.53**	88.93	81.56	51.83	93.59
7	92.91	76.34	95.23	91.12	92.06	85.04	92.55	**98.60**	87.39	86.30	94.82	94.68
8	**96.71**	95.08	89.36	98.10	88.87	88.72	95.62	94.37	87.76	78.81	79.50	95.92
9	57.56	48.11	46.65	30.64	26.85	34.78	44.24	**96.23**	31.80	49.65	9.190	62.64
10	96.91	98.02	98.24	98.07	96.29	95.90	98.50	92.41	97.11	97.79	93.87	**98.64**
OA	94.40	92.86	93.31	91.99	90.31	90.30	94.06	95.22	91.12	89.80	78.56	**95.41**
AA	94.42	92.67	93.73	92.21	90.21	87.02	94.18	95.80	90.00	89.13	69.53	**95.90**
Kappa	91.78	89.56	90.23	88.33	85.93	85.80	91.30	93.08	87.05	85.21	69.12	**93.25**

mechanism enabled the model to identify land-cover types within shadowed regions, achieving a high accuracy of 98.54% in classifying asphalt roads. In summary, SEDGM maintained a stable classification performance even under the interference of shadows or small targets.

3) Results for QUH-Pingan Dataset: The QUH-Pingan dataset contains numerous small target objects, such as ships and cars, which effectively test the classification performance of the models on objects at different scales. As shown in Table VII, most methods performed excellently in large-scale categories (e.g., seawater, concrete buildings, and roads). SSFTT achieved classification accuracies of 99.85% and 95.23% for seawater and steel-structured houses, respectively; CVT attained accuracies of 96.09% and 96.71% for concrete buildings and dock construction sites, respectively; and Mamba and KAN also achieved competitive accuracies in these categories. However, for small-scale categories (e.g., ships and cars), other methods performed poorly, with accuracies below 40%. GraphMamba effectively extracts discriminative spatial features through its graph-structured representation of nodes and edges, achieving superior classification performance for small-scale categories. SEDGM not only excelled in large-scale objects but also achieved the highest accuracy in difficult-to-distinguish, small targets, at 84.6% and 62.64% accuracy, respectively. Fig. 18 magnifies the local regions containing the small targets. SpectralFormer, MambaHSI, 2-D_KAN, and 3-D_KAN exhibited significant misclassifications, mistaking ships for concrete buildings and cars. Although CVT, BS2T, and MiM have fewer misclassifications, they connect ships with clear contour information to a single area, thereby losing edge information. In contrast, SEDGM successfully extracted spatial multiscale information by introducing a hierarchical spatial scanning mechanism that not only accurately identifies ship categories but also preserves more edge details.

4) Results for WHU-Hi-HongHu Dataset: The WHU-Hi-HongHu dataset contains different varieties of the same crop type, and the spectral mixing phenomenon among these subspecies poses significant challenges for classification algorithms. As shown in Table VIII, most methods accurately identified cotton, with BS2T achieving the highest classification accuracy of up to 98.95%. However, for closely related cotton stalks, the performance of most classifiers declined to varying degrees, with 3-D_KAN achieving only 19.80% accuracy, whereas SEDGM attained an optimal classification accuracy of 76.36%. Furthermore, SEDGM classified the

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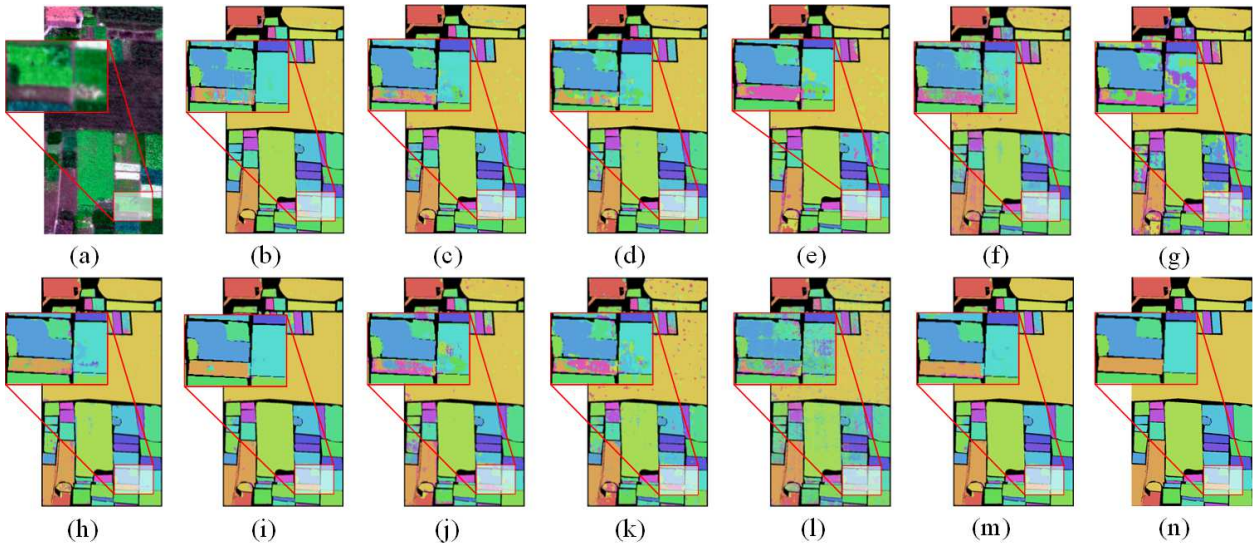


Fig. 19. Visual results of various methods on the WHU-Hi-HongHu dataset. The first row shows the classification results, and the second is an enlargement of the red box in the first row. (a) HSI. (b) CVT. (c) SpectralFormer. (d) SSFTT. (e) BS2T. (f) MambaHSI. (g) VIM. (h) MiM. (i) GraphMamba. (j) 1-D_KAN. (k) 2-D_KAN. (l) 3-D_KAN. (m) SEDGM. (n) Gt.

TABLE VIII
PERFORMANCE METRICS OF VARIOUS METHODS ON THE WHU-HI-HONGHU DATASET WITH OPTIMAL RESULTS IN BOLD

NO.	CVT	SpectralFormer	SSFTT	BS2T	MambaHSI	VIM	MiM	GraphMamba	1D_KAN	2D_KAN	3D_KAN	SEDGM
1	99.27	**99.52**	99.21	97.75	98.77	97.90	99.11	95.70	98.08	97.23	92.68	98.85
2	78.26	80.63	77.26	**98.95**	75.80	59.25	70.65	93.23	66.27	76.95	52.82	78.99
3	98.78	**98.94**	98.40	98.18	96.36	96.59	96.86	91.51	97.68	96.72	94.91	98.36
4	**99.69**	99.80	99.52	99.56	99.40	98.94	99.67	95.97	99.33	99.48	98.56	**99.76**
5	45.11	49.56	44.43	32.52	34.33	46.86	64.95	**96.57**	40.85	42.01	19.80	78.78
6	**98.77**	98.64	97.27	91.51	92.50	86.96	98.26	93.31	90.39	97.05	95.67	**99.46**
7	95.02	94.35	95.76	90.24	86.41	85.51	95.12	89.03	**96.74**	87.60	81.83	95.23
8	78.28	67.92	51.64	60.15	38.73	30.63	57.44	**93.12**	31.16	47.57	29.62	88.96
9	97.53	99.13	98.75	98.45	98.43	90.33	98.80	99.13	99.25	98.47	94.11	**99.59**
10	81.21	90.67	94.71	**99.82**	79.79	87.77	93.11	90.01	87.61	84.35	62.77	98.27
11	**93.83**	88.96	90.58	83.01	71.67	83.14	88.07	92.04	74.43	67.01	48.94	**94.23**
12	71.33	71.41	70.93	73.87	44.40	77.65	68.08	82.72	57.73	66.28	34.16	**87.43**
13	85.30	82.14	84.07	81.50	68.86	64.95	85.37	89.87	91.01	76.86	57.36	**95.41**
14	94.87	91.58	89.11	92.44	71.22	30.98	93.39	**97.81**	76.26	78.81	67.12	96.50
15	69.47	91.38	92.19	92.71	73.91	26.40	85.20	**100.0**	72.56	73.30	32.79	81.70
16	98.91	98.08	**99.70**	99.62	95.71	83.97	97.88	97.98	97.89	98.67	91.10	98.22
17	45.3 0	89.68	93.03	87.48	68.20	64.06	85.19	98.63	82.75	79.23	71.91	**96.49**
18	**91.05**	84.19	88.79	89.33	54.45	36.59	81.42	99.58	48.29	74.32	58.27	89.26
19	88.70	90.32	77.68	88.97	78.53	72.98	91.31	94.69	76.20	80.53	70.60	**96.87**
20	86.41	76.82	91.48	65.14	50.36	45.07	83.79	**96.37**	51.26	47.14	28.83	95.27
21	94.37	65.88	88.78	13.37	27.01	15.95	67.90	**99.92**	29.97	53.50	21.99	88.85
22	93.20	73.23	79.94	75.40	38.60	56.69	75.05	**99.34**	49.29	65.57	28.95	81.42
OA	90.62	91.16	90.51	86.77	82.18	79.91	91.63	94.20	84.79	86.25	73.66	**95.10**
AA	90.41	92.18	91.41	86.30	80.93	75.61	91.96	94.84	83.02	87.29	73.28	**95.79**
Kappa	88.32	88.98	88.18	83.59	78.03	74.95	89.53	92.72	81.09	83.01	68.23	**93.85**

subspecies of *Brassica chinensis* and small *Brassica chinensis* with accuracies exceeding 90%, ranking first among all methods. Fig. 19 shows the classification results of the different methods. In the regions marked by red boxes in the figure, the comparative methods exhibit significant misclassifications. Specifically, BS2T, MiM, and 2-D_KAN misclassified large areas of bare soil as broad bean; SSFTT and 3-D_KAN misidentified bare soil as Chinese cabbage and *Lactuca sativa*; and MambaHSI and VIM generally misclassified *Brassica chinensis* as carrot, Chinese cabbage, and *Lactuca sativa*. In contrast, SEDGM achieved better results in handling categories with high spectral mixing, demonstrating the highest classification accuracy and visual performance.

C. Discussion

1) Ablation Study for the Structure of SEDGM: To validate the effectiveness of the proposed modules, we conducted

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TABLE IX
CLASSIFICATION ACCURACIES OF SEDGM WITH DIFFERENT STRUCTURE ON FOUR DATASETS

Method	HGM	RSGM	SOFRM	Tangdaowan			Qingyun			Pingan			HongHu		
				OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
(a)SEDGM w/o RSGM and SOFRM	✓	-	-	94.04	96.59	93.26	88.06	89.75	84.45	93.47	93.75	90.44	84.82	81.67	81.19
(b)SEDGM w/o HGM and SOFRM	-	✓	-	83.46	89.24	81.46	89.14	90.64	85.84	89.69	89.50	85.05	92.87	92.82	91.04
(c)SEDGM w/o RSGM	✓	-	✓	96.19	97.75	95.68	91.06	93.07	88.31	94.26	94.77	91.58	93.44	94.01	91.78
(d)SEDGM w/o HGM	-	✓	✓	95.81	97.75	95.26	89.83	91.62	86.71	94.13	94.85	91.41	91.13	91.27	88.93
(e)SEDGM w/o SOFRM	✓	✓	-	93.93	96.36	93.13	88.71	90.84	85.25	93.21	93.27	90.05	93.42	93.79	91.74
(f)SEDGM	✓	✓	✓	**96.80**	**98.11**	**96.37**	**93.30**	**94.94**	**91.21**	**95.41**	**95.90**	**93.25**	**95.10**	**95.79**	**93.85**

TABLE X
CLASSIFICATION ACCURACIES OF SEDGM WITH DIFFERENT SPATIAL SCANNING METHODS ON FOUR DATASETS

Scan Scale	Tangdaowan			Qingyun			Pingan			HongHu		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
(a) Single scale	95.37	97.57	94.76	93.09	94.78	90.94	94.70	94.92	92.21	94.74	95.34	93.39
(b) Dual scale	96.55	97.92	96.09	93.16	94.91	91.03	95.31	95.86	93.11	94.80	95.24	93.47
(c) three-scale	96.8	98.11	96.37	93.3	94.94	91.21	95.41	95.9	93.25	95.1	95.79	93.85
(d) multi-scale	96.92	98.12	96.51	93.28	95.04	91.19	95.43	95.99	93.28	95.49	96.03	94.34

TABLE XI
CLASSIFICATION ACCURACIES OF SEDGM WITH DIFFERENT SPECTRAL SCANNING METHODS ON FOUR DATASETS

Scan Direction	Tangdaowan			Qingyun			Pingan			HongHu		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
(a) unidirectional	94.86	95.96	94.17	92.03	93.47	89.55	93.37	94.53	90.34	93.69	94.00	92.09
(b) bidirectional	95.07	97.32	94.42	92.78	94.53	90.54	94.12	94.48	91.37	92.76	93.70	90.95
(c) bidirectional and single shuffle	95.54	97.24	94.94	93.07	94.88	90.91	94.19	94.38	91.48	93.48	94.29	91.84
(d) bidirectional and double shuffle	96.8	98.11	96.37	93.3	94.94	91.21	95.41	95.9	93.25	95.1	95.79	93.85

ablation experiments on the model structure across four datasets. Table IX presents the classification accuracies of different network structures. The classification performance of the network declined when any module was removed. Furthermore, any combination of two modules demonstrated superior performance compared to any single module, whereas the integration of all modules most significantly enhanced network performance. Specifically, the HGM generates pixel sequences at different viewing distances and performs multidirectional scanning, deeply exploring the intrinsic connections and relational patterns within a sequence's spatial information, thereby achieving a richer representation of spatial features. This advantage is particularly pronounced in the Tangdaowan and Pingan datasets, where the structure incorporating the HGM spatial branch achieves a higher classification accuracy on both datasets. By randomly shuffling and rescanning the spectral data, RSGM successfully captures spectral features with a higher discriminative power that might be overlooked or weakened in traditional sequential scanning. This characteristic enables RSGM to excel in the Qingyun and HongHu datasets. The nonlinear analysis capability of the module for spectra allows it to effectively address spectral confusion issues caused by shadowed areas and subspecies of ground objects. In addition, SOFRM treats spectral information as a dynamically adjusted balancing factor, enabling flexible fusion with spatial features based on changes in spectral characteristics. This dynamic fusion approach allows SOFRM to adapt to the varying demands of different datasets for spatial and spectral information. The experimental results showed that incorporating SOFRM into any branch further enhances the classification accuracy. In summary, the inclusion of HGM, RSGM, and SOFRM significantly improved the classification performance of SEDGM across all tested datasets. The specific advantages demonstrated by each module on different datasets fully validate their adaptability and effectiveness in various application scenarios.

2) Comparison of Scanning Methods: To comprehensively validate the effectiveness of the proposed scanning method, we systematically compared the impact of different scanning strategies on the classification performance in both the spatial and spectral dimensions. Specifically, Table X proves that through scanning at different scales, we obtained a series of hierarchical sequences that not only captured detailed information in the images but also encompassed a broader spatial context, thereby effectively improving the model's ability to recognize spatial features. In the spectral dimension, the experimental results in Table XI show that unidirectional scanning methods have certain limitations in terms of classification accuracy, making it difficult to fully exploit the potential value of spectral information. However,

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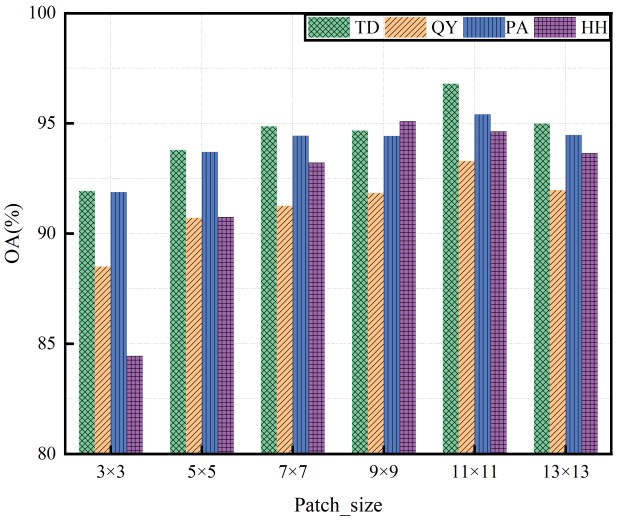


Fig. 20. OA(%) for different input patch sizes, where TD, QY, PA, and HH are QUH-Tangdaowan, QUH-Qingyun, QUH-Pingan, and WHI-Hi-HongHu, respectively.

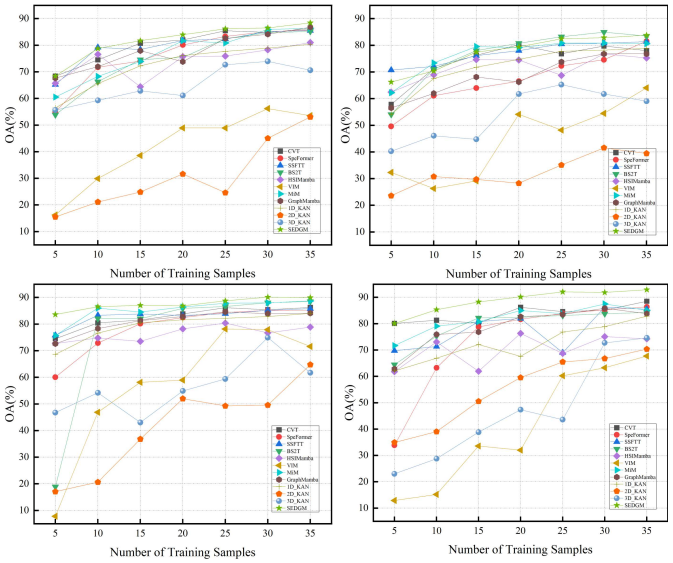


Fig. 21. Small sample learning ability of each method on four datasets. (a) QUH-Tangdaowan. (b) QUH-Qingyun. (c) QUH-Pingan. (d) WHI-Hi-HongHu.

when a shuffled path was introduced, the composition of the scanning sequences became more flexible and diverse. By shuffling, the model can capture more interdependencies and complementary information among the bands, thereby extracting more discriminative spectral features. In summary, the strategy combining the hierarchical and shuffled scanning mechanisms proposed in this article not only achieves the fusion of hierarchical features in the spatial dimension but also explores the nonlinear relationships between bands through the shuffled path in the spectral dimension, thereby significantly enhancing classification performance.

3) Discussion on the Input Patch Size: To explore the specific impact of the HSI input size on SEDGM performance, we conducted a comparative analysis of classification effectiveness across different input patch sizes on the four datasets. In the spectral Mamba branch, the model focuses on the scanning analysis along the spectral dimension, whereas in the spatial Mamba branch, it performs spatial scanning operations without altering the feature dimensions, enabling our method to accommodate inputs of various sizes. Fig. 20 shows the classification performance of these four datasets with the input patch size increasing from 3×3 to 13×13. The classification performance initially improved with increasing spatial size and subsequently declined. This can be attributed to the fact that smaller neighborhood sizes often limit the model's receptive field, resulting in insufficient information capture, whereas larger patches can encompass more spatial information, thereby enhancing the classification performance to some extent. However, as the patch size increases further, it introduces more potentially interfering pixels, particularly at category boundaries, which can lead to a decline in classification accuracy. Notably, this trend was not entirely consistent across the QUH-Tangdaowan, QUH-Qingyun, QUH-Pingan, and WHI-Hi-HongHu datasets. Specifically, for the QUH-Tangdaowan, QUH-Qingyun, and QUH-Pingan datasets, the model achieved optimal performance when the subcube size was 11×11, whereas for the WHI-Hi-HongHu dataset, the best performance was observed at a size of 9×9.

4) Sensitivity in Relation to the Training Samples: We evaluated the learning capabilities of different methods with limited samples and analyzed the robustness and generalization ability of the models with varying numbers of training samples. Specifically, we randomly selected 5, 10, 15, 20, 25, 30, and 35 samples from each category for training and plotted the OA curves for each method across the four datasets, as shown in Fig. 21. As the number of training samples increased, the performance of all methods showed an upward trend, primarily because of the richer informational foundation provided by the more labeled samples for the classification task. Notably, although our method did not achieve the highest classification accuracy across all sample sizes on every dataset (as illustrated by the Qingyun dataset), it consistently maintained the highest level of accuracy in the overall performance. This is primarily attributed to the innovative dual-branch Mamba structure design, which fully exploits the spatial–spectral information of the samples. Through the introduction of a hierarchical spatial scanning mechanism within the Mamba structure, the spatial branch is capable of deeply mining multiscale contextual information and constructing a dynamic spectral analysis framework that effectively extracts nonlinear relationships between spectra. Furthermore, both branches incorporate a distance-oriented dynamic discrimination gate, which enables the precise selection of more discriminative features. Additionally, SOFRM facilitates deep review and fine-tuning of features, further perceiving and extracting higher dimensional knowledge information, demonstrating a more stable classification performance across different datasets and varying numbers of training samples. These observations indicate that our approach has excellent few-shot classification capabilities.

5) Robustness to Spectral Corruption and Dataset-Specific Shifts: To comprehensively assess the model's robustness to

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TABLE XII
ROBUSTNESS EVALUATION AGAINST SPECTRAL DROPOUT (RATES: 0.1/0.3/0.5)

Mask Ratio	Tangdaowan			Qingyun			Pingan			HongHu		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
0.1	94.46	97.31	93.74	95.23	95.70	9300	92.27	94.06	89.87	94.77	95.12	93.44
0.3	93.27	95.25	92.39	94.89	95.33	92.51	90.17	91.82	87.16	93.63	94.72	92.05
0.5	84.90	87.42	84.24	85.08	85.42	82.78	89.59	91.79	86.38	81.64	81.52	77.52

TABLE XIII
PERFORMANCE EVALUATION UNDER SPECTRAL SHIFTS

Method	Tangdaowan			Qingyun			Pingan			HongHu		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
A	94.96	96.60	93.43	88.25	90.88	84.69	93.95	94.60	91.15	92.50	94.35	90.86
B	96.47	98.05	96.00	92.73	94.31	90.46	95.18	95.79	92.93	94.19	95.15	92.73
C	96.80	98.11	96.37	93.30	94.94	91.21	95.41	95.90	93.25	95.10	95.79	93.85

TABLE XIV
EFFICIENCY COMPARISON OF DIFFERENT METHODS (BOLD: BEST, UNDERLINE: SECOND BEST)

Method	Tangdaowan			Qingyun			Pingan			HongHu		
	Params(M)	Flops(M)	Test Time(s)	Params(M)	Flops(M)	Test Time(s)	Params(M)	Flops(M)	Test Time(s)	Params(M)	Flops(M)	Test Time(s)
CVT	17.60	2575.36	3337.68	17.84	2576.16	3152.01	17.74	2575.83	1909.61	17.84	1787.90	899.30
SpecFormer	0.62	89.55	**186.40**	0.86	146.93	**167.65**	0.61	89.68	**168.52**	0.83	138.31	**86.04**
SSFTT	2.62	129.38	391.02	3.93	198.72	214.01	2.62	129.38	239.41	3.92	103.42	176.52
BS2T	0.33	141.97	331.10	0.49	219.08	562.46	0.33	141.96	297.36	0.49	146.66	160.44
MambaHSI	13.43	**13.87**	193.14	19.26	19.72	172.84	19.27	19.73	187.64	13.73	14.16	158.21
VIM	0.38	17.83	343.72	0.38	**19.34**	269.59	0.35	**17.83**	282.92	0.37	**13.45**	140.64
MiM	**0.18**	141.66	962.83	**0.18**	144.75	1051.92	**0.18**	141.67	1286.90	**0.18**	103.47	422.97
GraphMamba	1.38	180.99	-	1.38	183.50	-	1.36	180.71	-	1.32	116.56	-
SEDGM	0.43	43.86	422.00	0.43	43.86	280.27	0.42	43.85	357.95	0.41	34.93	124.22

* Since GraphMamba requires constructing graph structures during inference, leading to significantly increased memory usage and computation time, this study does not include it in the comparative analysis.

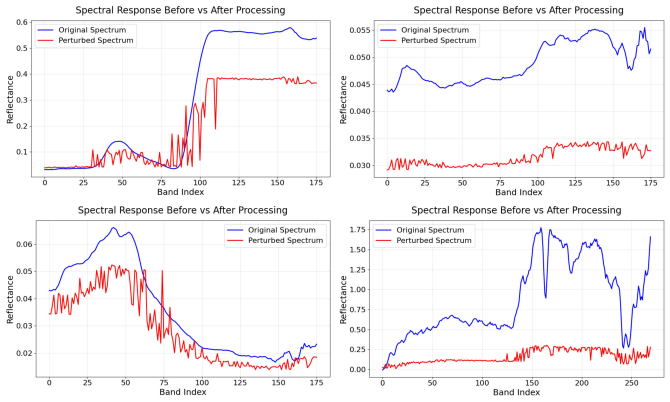


Fig. 22. Spectral curves of central pixels before and after spectral distortion for four datasets. (a) Tangdaowan. (b) Qingyun. (c) Pingan. (d) HongHu (blue: original spectra, red: shifts spectra).

spectral anomalies (information loss and distribution shifts), we designed two systematic experiments.

1) *Spectral Dropout:* Random masking was applied along the spectral dimension (dropout rates: 0.1, 0.3, 0.5) to simulate sensor failures or atmospheric interference.
2) *Spectral Shift:* A parameterized Gaussian kernel model was employed to simulate spectral distortion caused by sensor aging. By introducing random center-wavelength shifts and bandwidth perturbations, we constructed datasets with significant spectral curve variations. Fig. 22(a)–(d) illustrates the spectral curves of central pixels before and after spectral shifts for the Tangdaowan, Qingyun, Pingan, and HongHu datasets, respectively. As shown in Table XII, when the spectral dropout ratio was 0.1 or 0.3, the model maintained competitive classification performance. However, once the dropout rate increased to 0.5, excessive band masking led to critical information loss, impairing the model's ability to distinguish between classes accurately. Notably, the classification accuracy dropped more significantly on the HongHu dataset. We attribute this to HongHu's large number of classes and high spectral confusion, making it difficult for the model to extract sufficient discriminative features under severe spectral information loss. Table XIII presents the classification accuracy of our method on spectrally shifted datasets. Specifically, Group A: SEDGM without the SOFRM trained on spectrally shifts data. Group B: Full SEDGM (with SOFRM) trained on spectrally shifts data. Group C: Full SEDGM trained on the original dataset. The results clearly indicate performance degradation after spectral distortion, particularly in the Qingyun and

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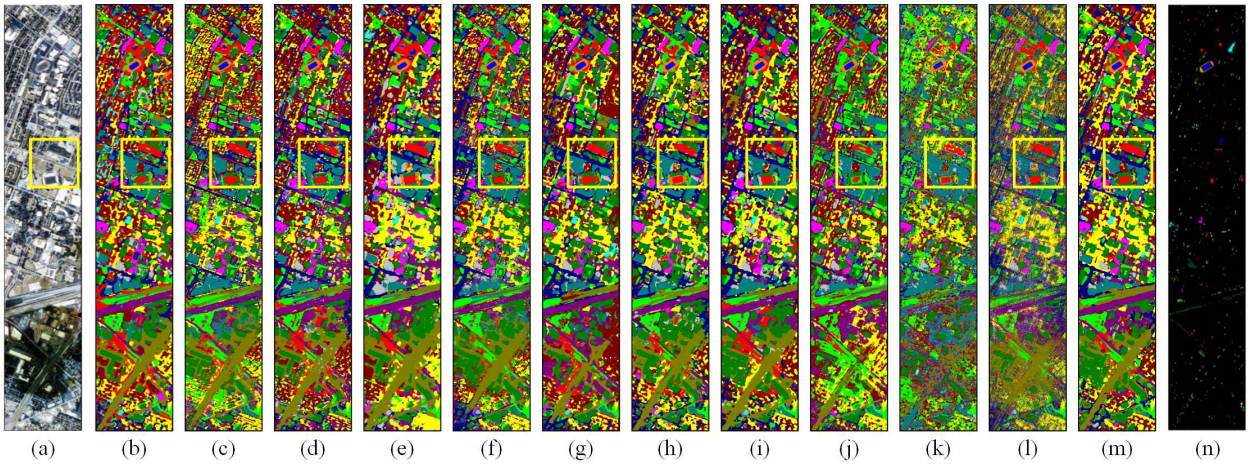


Fig. 23. Visual results of various methods on the Houston 2013 dataset. The first row shows the classification results, and the second is an enlargement of the red box in the first row. (a) HSI. (b) CVT. (c) SpectralFormer. (d) SSFTT. (e) BS2T. (f) MambaHSI. (g) VIM. (h) MiM. (i) GraphMamba. (j) 1-D_KAN. (k) 2-D_KAN. (l) 3-D_KAN. (m) SEDGM. (n) Gt.

TABLE XV
PERFORMANCE METRICS OF VARIOUS METHODS ON THE HOUSTON 2013 DATASET WITH OPTIMAL RESULTS IN BOLD

NO.	CVT	SpeFormer	SSFTT	BS2T	MambaHSI	VIM	MiM	GraphMamba	1D_KAN	2D_KAN	3D_KAN	SEDGM
1	92.36	96.27	**99.09**	90.02	95.61	91.88	95.37	98.03	91.57	83.52	89.29	92.52
2	97.76	92.06	**98.31**	96.17	89.76	97.54	94.66	96.41	92.86	63.00	81.74	97.16
3	100.0	99.70	100.0	100.0	100.0	98.96	100.0	100.0	100.0	98.56	99.40	**100.0**
4	91.87	94..22	**99.26**	82.98	94.25	93.79	92.55	99.92	92.16	68.26	80.56	94.11
5	95.39	98.85	98.85	98.85	99.67	100.0	99.92	100.0	93.25	95.38	95.18	**100.0**
6	71.26	100.0	**100.0**	100.0	92.48	91.61	80.62	100.0	100.0	74.20	86.01	99.66
7	83.00	95.00	92.41	87.22	89.95	89.26	92.79	87.00	92.21	64.19	77.00	**98.70**
8	96.83	91.99	**100.0**	98.48	95.58	97.77	86.86	86.24	85.80	72.52	79.57	98.69
9	98.98	94.51	90.36	**99.09**	81.90	91.07	89.64	87.07	77.49	60.88	51.65	93.37
10	88.01	90.08	95.30	84.05	81.00	85.45	80.97	91.65	78.81	45.90	64.55	**98.60**
11	96.01	96.94	89.06	**97.13**	95.52	86.10	97.00	99.75	86.50	65.97	67.99	96.84
12	92.48	85.21	96.84	91.91	73.48	91.35	90..57	88.69	84.54	45.10	69.76	**97.22**
13	91.74	93.22	94.16	46.24	78.06	84.16	8473	89.75	76.38	79.76	66.50	**99.31**
14	99.00	91.92	100.0	100.0	88.84	100.0	100.0	96.23	95.67	98.73	83.90	**100.0**
15	96.18	92.11	99.21	96.48	98.28	95.67	90.00	**100.0**	96.33	90.91	96.04	95.45
OA	92.60	93.33	95.92	89.79	89.36	92.21	91.45	94.14	88.25	68.34	77.01	**96.65**
AA	93.83	93.49	96.57	91.54	91.04	93.42	92.57	94.72	89.50	72.66	80.40	**97.28**
Kappa	92.01	92.80	95.59	88.98	88.50	91.59	90.76	93.67	87.30	65.78	75.17	**96.37**

HongHu datasets, where spectral confusion is more pronounced. Nevertheless, the model still delivered competitive classification performance despite spectral perturbations. Notably, the full SEDGM achieved near-original performance, demonstrating its ability to extract highly discriminative features under spectral shifts and maintain precise classification. This strongly validates the robustness of SEDGM in handling spectral variations.

6) Comparison of Runtime and Model Complexity: We have conducted comprehensive statistics on the number of parameters, FLOPs, and inference time for all algorithms. In terms of experimental setup, except for the Honghu dataset where the patch size was set to 9, the patch size for the other three datasets was uniformly set to 11, with a batch size of 1. To ensure the fairness and comparability of the experiments, all algorithms were trained and tested using input data of identical dimensions under identical hardware and software environments. Table XIV provides a detailed summary of the specific values for the number of parameters, FLOPs, and inference time of each algorithm across the three datasets. Analysis of the table reveals that the inference time of the SEDGM model falls within a reasonable and acceptable range, posing no excessive time cost burden for practical applications. Moreover, compared to the baseline transformer and Mamba models, SEDGM demonstrates significant advantages in computational complexity, exhibiting relatively lower computational overhead. Therefore, combining its superior classification performance and lower training costs, SEDGM shows considerable potential for practical applications.

7) Classification Performance on Airborne Hyperspectral Datasets: To validate the classification performance of our method on spaceborne or airborne hyperspectral datasets, we conducted additional experiments on the Houston2013

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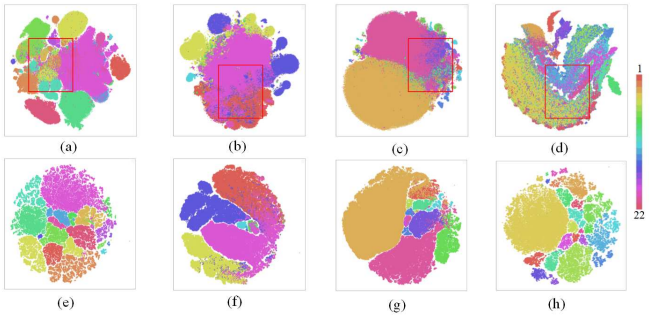


Fig. 24. Visualization of t-SNE features of SEDGM on five datasets. (a)–(d) Distribution of the original datasets. (e)–(h) Distribution of features extracted by SEDGM. Each sample of different categories is represented by color dots.

dataset. Comparative experiments with other methods were also performed on this dataset for comprehensive evaluation. The Houston 2013 dataset was released by the IEEE GRSS Data Fusion Competition Organizing Committee, and the original data was collected by the National Airborne Laser Mapping Center (NCALM) in the area around the University of Houston. The dataset space size is 1905 × 349, with a spectral range covering the visible light to near-infrared region, totaling 144 spectral channels. The ground resolution corresponding to each pixel is 2.5 m. The dataset provides 15 029 reference samples with accurate category labels, which are classified into 15 typical types of ground features. In this experiment, we randomly selected 30 samples from each category for training, accounting for 2% of the total sample size. The classification accuracies and visual results of all methods on the Houston 2013 dataset are shown in Table XV and Fig. 23, respectively. The experimental results demonstrate that the proposed method achieves superior classification performance on the airborne dataset, with OA, AA, and kappa all outperforming comparative methods. Notably, it attains the highest classification precision for typical land-cover categories including Residential, Highway, and Parking-lot-1. As visually confirmed in the yellow box regions of Fig. 23, comparative methods CVT, SSFTT, and 1-D_kan exhibit noticeable isolated misclassifications, while BS2T and 3-D_kan display edge blurring and boundary buffer effects. In contrast, the classification maps generated by our method show smoother homogeneous regions and sharper object boundaries, further validating its robustness and accuracy on airborne datasets.

8) Separability Visualization: To demonstrate the model's feature representation capability more intuitively, we employed t-SNE technology to map the original high-dimensional data of the hyperspectral images and the spectral–spatial features learned by the model into a two-dimensional space [61], as shown in Fig. 24. In the original HSI data, the sample distributions of various categories exhibited a highly mixed state across the four different datasets, with blurred boundaries between them, increasing the classification difficulty. However, after undergoing refined feature extraction using SEDGM, the samples became more compact and uniform within classes, with more distinct boundaries between classes. Samples with similar features formed tight intraclass clusters with significantly reduced internal distances, whereas samples from different categories were effectively separated, with markedly increased interclass distances. This distribution pattern of "like gathers with like, and unlike separates" greatly enhances the accuracy of subsequent classification tasks. Notably, as highlighted by the red boxes in the figure, certain sample regions that were difficult to identify or confused in the original data underwent significant enhancement in their intrinsic structure and differentiation after feature transformation using SEDGM. This transformation makes these previously indistinguishable samples easily distinguishable at the feature level, demonstrating the superior feature extraction capability of SEDGM.

V. Conclusion

This study proposed SEDGM to address the limitations of existing Mamba in hyperspectral image classification tasks. Through the collaborative design of spatial and spectral gating, SEDGM effectively resolved the issues of insufficient hierarchical spatial–contextual representation and difficulty in modeling nonlinear spectral interactions. HGM effectively mines multiscale spatial features and their intrinsic relationships in hyperspectral imagery, whereas RSGM successfully models covarying patterns across spectral dimensions, enhancing the discriminative power of spectral features. Additionally, SEDGM introduces a distance-oriented dynamic discrimination gate and a bias-aware attention mechanism, further compensating for the shortcomings of traditional Mamba in modeling structural information when processing sequence data. Through the dynamic adjustment of the SOFRM, SEDGM achieves efficient fusion of spatial and spectral information, thereby enhancing the model's generalization and feature representation capabilities. The performance of SEDGM was extensively evaluated on four of the latest and largest public datasets, showing significant advantages over other deep learning methods and promising valuable applications for HSIC tasks.

VI. Credit Authorship Contribution Statement

Shuai Zhang: data curation, methodology, writing—original draft preparation. Yonghua Jiang: methodology, writing—original draft preparation. Chengjun Wang: conceptualization of this study, software. Meilin Tan: data curation. Bin Du: data curation. Xin Shen: data curation.

VII. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Acknowledgment

The authors would like to thank the anonymous reviewers for their detailed and constructive comments and suggestions. They also thank the Supercomputing Center of Wuhan University for providing computing power to conduct experiments.

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